EVERY VALUATION RING IS F-SMOOTH

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Abstract

We prove that every valuation ring is F-smooth, in the sense of Bhatt and Mathew, at every prime p. Along the way we show that for a henselian valuation ring V of mixed characteristic $(0, p)$ and every $n \geq 0$, the mod-p syntomic-to-étale comparison

$$\mathbb{F}_p(n)_{\mathrm{syn}}(V) \longrightarrow R\Gamma_{\acute{e}t}\big(V[1/p], \mu_p^{\otimes n}\big)$$

is an isomorphism in degrees $< n$ and injective in degree n. The proof reduces to complete valuation rings of rank one. For these, weak F-smoothness follows from Mao's Frobenius–Witt cotangent complex, and what remains is the vanishing of certain Nygaard-completion defects. We prove this by induction on the cohomological degree. The induction uses two further ingredients: injectivity of the Galois symbol on improved Milnor K-theory modulo p, and a trivialization of the Breuil–Kisin line after a Frobenius-type base change.

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1 Introduction

1.1 Main theorems

Bhatt and Mathew introduced F-smoothness as a condition on p-quasisyntomic rings under which syntomic cohomology behaves as it does for smooth schemes [4]. Regular rings are F-smooth [4, Theorem 4.15], and so are Cartier smooth $\mathbf{F}_p$-algebras [4, Proposition 4.14]; this already covers valuation rings of characteristic p (Proposition 10.1). The purpose of this paper is to treat valuation rings of mixed characteristic.

For a p-complete ring R, write

$$S_n(R) = \mathbb{F}_p(n)_{\mathrm{syn}}(R), \qquad E_n(R) = R\Gamma_{\acute{e}t}(R[1/p], \mu_p^{\otimes n}),$$

and let $\gamma_{R,n} : S_n(R) \to E_n(R)$ be the natural comparison. For a valuation ring that is not p-complete we define syntomic cohomology through the completion and the generic fiber, as in [1].

Theorem 1.1. *Let V be a henselian valuation ring of mixed characteristic $(0,p)$. For every $n \geq 0$,*

$$\mathrm{cofib}(S_n(V) \xrightarrow{\gamma_{V,n}} E_n(V)) \in D^{\geq n}(\mathbb{F}_p).$$

Equivalently, $\gamma_{V,n}$ is an isomorphism on H^q for $q < n$ and injective on H^n.

No assumption is made on the rank or the value group of V, and $p = 2$ is allowed.

Theorem 1.2. *Every valuation ring is F-smooth at every prime p, in the sense of [4, Definition 4.1].*

Theorem 1.1 verifies the hypothesis $\mathrm{Val}(\operatorname{Spec}\mathbf{Z}, p, j)$ of Bachmann–Elmanto–Morrow [1] for every j, and Theorem 1.2 says that every ring is cdh-locally F-smooth in the sense of Bouis–Kundu [10]. Some consequences for motivic cohomology are collected in Section 12.

1.2 Outline of the proofs

The proof of Theorem 1.1 begins with the valuative reduction of Bachmann–Elmanto–Morrow, which reduces it to complete valuation rings of rank one (Section 2). For these, surjectivity below the weight is obtained by lifting generic Milnor symbols to regular semistable local models (Proposition 2.6). What remains is injectivity.

The first part of the argument is prismatic. The Nygaard–Sen fiber sequence and Mao's Frobenius–Witt cotangent complex give a Tor-amplitude bound for the cofiber of multiplication by the Nygaard class θ (Section 3); this is weak F-smoothness in rank one. We then compare the twisted Nygaard filtration on Hodge–Tate cohomology with its completion. Finite Bott strictness relates the completion defect E_k to the cofiber of Bott multiplication on syntomic cohomology (Section 4), and a separation argument on the first nontrivial term of a perfectoid Čech nerve gives $E_k \in D^{\geq 1}$ (Section 5).

The second ingredient is algebraic: for nondiscrete V of rank one,

$$\widehat{K}^M_n(V)/p \hookrightarrow K^M_n(\operatorname{Frac} V)/p \qquad (n \geq 1).$$

The proof (Sections 6–7) uses the fiber product $V \times_k V$ over the residue field to construct operations on relative Milnor K-groups.

The third ingredient is geometric (Section 8). Let $\psi : V \to V/p$, $v \mapsto \bar{v}^p$. Using the Hodge–Tate stack and the prism $(\mathbf{Z}_p[[T, B]], (p - T^p B))$, we trivialize the pullback of the Breuil–Kisin line along ψ, so that all the $\psi^* E_k$ are equivalent. Together with the endpoint and Bott localization, this lets Section 9 prove $E_k = 0$ for every k by induction. This settles complete rank one, and excision gives Theorem 1.1 in general (Section 11).

For Theorem 1.2 we apply excision separately to the fibers of θ and to the completion defects, and use a localization lemma for Tor-amplitude to pass to arbitrary rank (Section 10). Section 12 records applications.

1.3 Conventions

We follow Bhatt–Lurie [2] for prismatic and syntomic cohomology, Bhatt–Scholze [7] for Hodge–Tate and Nygaard calculations, and Bhatt–Mathew [4] for Bott classes and F-smoothness. Tensor products, completions, fibers and cofibers are derived unless stated otherwise. Complexes are cohomologically graded, so $M[-r]$ lies in degree r for a discrete module M.

For a p-quasisyntomic ring R, $\mathbb{\Delta}_R$ denotes absolute prismatic cohomology, $\mathcal{N}^{\geq r}\mathbb{\Delta}_R$ its Nygaard filtration, and $\mathbb{\Delta}_R\{k\}$ the kth Breuil–Kisin twist. A bar denotes Hodge–Tate specialization: $\overline{\mathbb{\Delta}}_R\{k\}$ is the reduction of $\mathbb{\Delta}_R\{k\}$ along the Hodge–Tate divisor. A hat on a prismatic complex denotes Nygaard completion, and $\widehat{H}_k(R)$ is the completion of $\overline{\mathbb{\Delta}}_R\{k\}/p$ for the induced twisted Nygaard filtration (Section 4.2). The degree-p Nygaard class is θ, and the mod-p syntomic Bott class is v_1, of twist $p - 1$.

2 Reduction to complete rank one and surjectivity below the weight

2.1 The valuative reduction

The following reduction is the form of [1, Remark 8.4] used here; its proof rests on the v-excision square of [1, Proposition 7.25].

Proposition 2.1 (Bachmann–Elmanto–Morrow)**.** *Fix p and n. If*

$$\operatorname{cofib}(S_n(W) \to E_n(W)) \in D^{\geq n}$$

for every p-complete henselian rank-one valuation ring W of mixed characteristic, the same holds for every henselian mixed-characteristic valuation ring.

Proof. We sketch the argument. Completion does not change the comparison cofiber, by the completion/generic-fiber definition of syntomic cohomology. For a complete valuation ring V, put $\mathfrak{q} = \sqrt{pV}$. The localization $V_{\mathfrak{q}}$ has rank one and the same p-inverted fraction field. The v-excision square compares V, $V_{\mathfrak{q}}$, $V/\mathfrak{q}$ and $\kappa(\mathfrak{q})$. The characteristic-p terms have comparison cofibers in $D^{\geq n}$; in degree n this uses injectivity of logarithmic forms from a Cartier-smooth valuation ring to its fraction field. Hence connectivity transfers from $V_{\mathfrak{q}}$ to V. See [1, Proposition 7.25 and Remark 8.4]. □

Throughout Sections 2–9, V is a p-complete henselian rank-one valuation ring of mixed characteristic with fraction field K, unless specified otherwise.

2.2 A density lemma

Lemma 2.2. *Let $\Gamma \subset \mathbb{R}$ be a nondiscrete subgroup and $m \geq 1$. Every coset of $m\Gamma$ is dense in $\mathbb{R}$. Consequently, if K is a rank-one valued field with nondiscrete value group, every class of $K^\times/K^{\times m}$ has representatives of arbitrarily small positive valuation.*

Proof. $m\Gamma$ is again a nondiscrete subgroup of $\mathbb{R}$, hence dense, and so are its cosets. Given $z \in K^\times$, choose $\gamma \in \Gamma$ with $v(z) + m\gamma$ in the required interval and $h \in K^\times$ with $v(h) = \gamma$; then zh^m works. □

2.3 The perfectoid quasisyntomic cover

We will use the following cover throughout.

Lemma 2.3 (Perfectoid valuation cover)**.** *Let C be the completion of an algebraic closure of K, and put $W = \mathcal{O}_C$. The map $V \to W$ is a p-completely faithfully flat quasisyntomic cover. For every $a \geq 0$,*

$$A_a = \left(W^{\otimes_V (a+1)}\right)^\wedge_p$$

is a discrete p-torsion-free quasiregular semiperfectoid ring. Absolute prismatic cohomology, its finite Nygaard quotients and Hodge–Tate specializations, and syntomic cohomology satisfy quasisyntomic descent on the resulting Čech nerve.

Proof. For an extension of valued fields, the cotangent complex of the extension of valuation rings has no homology above degree one, and its degree-one homology is torsion-free [11, Theorem 6.5.8]. Since torsion-free modules over a valuation domain are flat and every module over a valuation domain has flat dimension at most one, it gives the quasisyntomic Tor bound for $V \to W$. The valuation extension is faithfully flat, and p-adic completion preserves its reductions modulo p^r.

The tensor powers are flat over V, hence p-torsion-free; their derived p-completions are discrete and remain p-torsion-free. Quasisyntomicity is preserved by the relevant base changes. To see semiperfectoidness, choose compatible p-power roots in the algebraically closed perfectoid valuation ring W of elements in the additional tensor factors. A perfectoid root-polynomial algebra over the first copy of W surjects modulo p onto the tensor product; successive p-adic lifting gives a surjection onto its completion. Thus every A_a is qrsp.

Quasisyntomic descent follows from [7, Construction 7.6 and Theorem 12.2] and [2, Section 5.5]. Syntomic descent then follows from its description as the Frobenius-minus-canonical-map fiber [2, Proposition 7.4.6]. □

2.4 Prime-to-p cyclotomic descent

We will repeatedly adjoin μ_p. The integral closure of V in $K(\mu_p)$ need not be finite over V, but descent can still be carried out with traces on finite free subalgebras.

Lemma 2.4. *Let V be henselian of rank one, let L/K be cyclic of degree m prime to p, and suppose $\mu_m \subset K$. If V' is the valuation ring of L, there is a finite étale valuation extension U/V such that V' is a filtered union of finite free U-algebras of the form $U[t]$, each with fraction field L. Their rank over V is m.*

Proof. Choose a Kummer generator r with $r^m \in K$. Let e be the order of $v(r)$ in Γ_L/Γ_K and put $f = m/e$. After multiplying r^e by a scalar in $K^\times$, obtain a unit s satisfying $s^f \in V^\times$. Since f is

prime to p and s^f is a unit, $U = V[s]$ is finite étale of rank f over V, hence normal. It is integral over V with fraction field $K(s)$, so it is the integral closure of V in $K(s)$, which is the valuation ring of $K(s)$ because V is henselian.

Now $L = \mathrm{Frac}(U)(r)$ has degree e, and the values of $1, r, \dots, r^{e-1}$ are distinct modulo Γ_K. In the nondiscrete case, replace r by br with $b \in K(s)^\times$ and $v(br) > 0$ arbitrarily small. Any prescribed finite subset of V' lies in the algebra $U[br]$, and these algebras form a directed system. In the discrete case choose a Kummer generator of minimal positive value and obtain $V' = U[r]$. □

Let $B = A[T]/(f)$ with f monic of degree d. The homogenization of f defines a relative effective Cartier divisor $i : \operatorname{Spec} B \hookrightarrow \mathbf{P}^1_A$. The codimension-one Gysin map and the projective bundle formula give

$$S_i(B) \longrightarrow S_{i+1}(\mathbf{P}^1_A)[2] \longrightarrow S_i(A),$$

where the second arrow projects to the coefficient of $c_1(\mathcal{O}(1))$. Denote the composite by Tr_f. Here we use the Gysin construction of [20, Definition 5.32] and the projective bundle formula [2, Theorem 9.1.1]. The divisor has class $c_1(\mathcal{O}(d)) = d\, c_1(\mathcal{O}(1))$, so the projection formula gives $\mathrm{Tr}_f \circ \mathrm{res} = d$. On a separable generic fiber the corresponding étale Gysin map is the usual finite transfer. We use these properties only for the finite free polynomial quotients in Lemma 2.4.

Proposition 2.5 (Cyclotomic descent)**.** *Let $L = K(\mu_p)$ with valuation ring V'. For every i, q:*

(1) $H^q S_i(V) \to H^q S_i(V')$ is injective;

(2) an étale class over K whose restriction to L has a syntomic lift over V' has a syntomic lift over V;

(3) injectivity of $H^q S_i(V') \to H^q E_i(V')$ implies injectivity over V.

Proof. The degree $m = [L : K]$ divides $p - 1$. The m distinct roots of $X^m - 1$ in $\mathbf{F}_p$ lift to V by henselianity, so $\mu_m \subset K$, as required in Lemma 2.4. By Lemma 2.4, V' is a filtered union of finite free polynomial quotients B_t of total rank m over V, after a finite étale intermediate extension. Mod-p syntomic cohomology commutes with filtered colimits [2, Corollary 8.4.11]. The traces give compatible maps

$$T_t : H^q S_i(B_t) \longrightarrow H^q S_i(V), \qquad T_t \circ \mathrm{res} = m,$$

compatible with generic corestriction. Since $m \in \mathbf{F}_p^\times$, (1) follows. If α_t lifts the restriction of an étale class β, then

$$\gamma(T_t \alpha_t) = \mathrm{Cor}_{L/K} \mathrm{Res}_{L/K}(\beta) = m\beta,$$

which proves (2) after multiplying by m^{-1}. Part (3) follows from (1). For $p = 2$, $m = 1$. □

2.5 Semistable lifting below the weight

Proposition 2.6 (Surjectivity below the weight)**.** *For every $i \geq 1$ and $0 \leq q < i$, the comparison*

$$H^q S_i(V) \longrightarrow H^q_{\acute{e}t}(K, \mu_p^{\otimes i})$$

is surjective.

Proof. By Proposition 2.5 we may assume $\mu_p \subset K$ and fix a primitive root ζ. If the valuation is discrete, V is a complete DVR and the assertion follows from the regular comparison theorem [4, Theorem 4.15 and Proposition 5.2]. Assume the value group is nondiscrete and put $\lambda = \zeta - 1$. By the norm-residue theorem [22, Theorem 6.16], the target is generated by classes

$$\zeta^{\otimes(i-q)} \cup [a_1] \cup \cdots \cup [a_q].$$

By Lemma 2.2, replace each a_r in its pth-power class so that

$$0 < v(a_r) < \frac{v(\lambda)}{q+1}.$$

Then $d = \lambda/(a_1 \cdots a_q)$ lies in the maximal ideal. Consider the henselian regular local ring

$$R = (\mathbf{Z}_p[\zeta][X_1, \ldots, X_q, Y]/(X_1 \cdots X_q Y - \lambda))^h_{(X_1,\ldots,X_q,Y)}.$$

It is regular of dimension $q+1$ and the assignments $X_r \mapsto a_r$, $Y \mapsto d$ define a local map $R \to V$. After inverting p, the X_r are units. Every regular noetherian ring is F-smooth, and on a p-torsion-free F-smooth ring the syntomic-to-étale comparison is an isomorphism below the weight [4, Theorem 4.15 and Proposition 5.2]. Since $q < i$, the generic symbol has a syntomic lift on R, whose pullback gives the desired lift on V. For $p = 2$, $\lambda = -2$. □

3 Weak F-smoothness of complete rank-one valuation rings

In this section we bound the Tor-amplitude of the cofiber of θ on Nygaard graded pieces, in all weights at once (Theorem 3.6). Discrete valuation rings are regular, so they are covered by [4]; the work is in the nondiscrete case.

3.1 The Bott-defect triangle

For an animated ring R, put

$$A_R = R \otimes^{\mathbf{L}}_{\mathbf{Z}} \mathbf{F}_p, \qquad C_R = L_{A_R/\mathbf{F}_p}, \qquad \mathscr{L}_n(R) = (\mathbf{L}\textstyle\bigwedge^n C_R)[-n].$$

Write

$$G_n(R) = \mathrm{gr}^n_{\mathrm{N}} \mathbb{\Delta}_R/p, \qquad Q_i(R) = \mathrm{cofib}(\theta : G_i(R) \to G_{i+p}(R)).$$

The Nygaard–Sen description of $\mathrm{gr}^n_{\mathrm{N}} \mathbb{\Delta}_R$ gives a fiber sequence

$$\mathrm{gr}^n_{\mathrm{N}} \mathbb{\Delta}_R \longrightarrow \mathrm{Fil}^{\mathrm{conj}}_n \widehat{\Omega}^{\mathrm{D}}_R \xrightarrow{\Theta+n} \mathrm{Fil}^{\mathrm{conj}}_{n-1} \widehat{\Omega}^{\mathrm{D}}_R,$$

where the nth conjugate graded piece is $(\bigwedge^n L_{R/\mathbf{Z}})^\wedge_p[-n]$ and Θ acts there by $-n$; see [2, Construction 4.7.1 and Remark 5.5.8]. We also use the polynomial-variable decompositions of [2, Example 5.6.10 and Variants 5.6.12–5.6.13].

Lemma 3.1. *For every animated ring R and $i \geq 0$ there is a natural fiber sequence in $D(A_R)$*

$$Q_i(R) \xrightarrow{q_i} \mathscr{L}_{i+p}(R) \xrightarrow{\beta_i} \mathscr{L}_i(R).$$

The construction is compatible with adjoining polynomial variables: under

$$\mathscr{L}_n(R[X]) \simeq \mathscr{L}_n(R)[X] \oplus \mathscr{L}_{n-1}(R)[X][-1],$$

the map $\beta_i(R[X])$ decomposes as $\beta_i(R)[X] \oplus \beta_{i-1}(R)[X][-1]$.

Proof. The multiplicative Hodge projection gives $h_n : G_n(R) \to \mathscr{L}_n(R)$. The Hodge image of θ is canonically zero, since it is pulled back from $\mathbf{Z}_p$ and $\mathscr{L}_p(\mathbf{Z}_p) = 0$. Thus h_{i+p} descends to $q_i : Q_i \to \mathscr{L}_{i+p}$. We keep track of this nullhomotopy: it is needed to animate the construction and to identify the scalars in Lemma 3.3.

Suppose first that R is a polynomial $\mathbf{Z}$-algebra. Its conjugate filtration is the Postnikov filtration. Writing $\widehat{\Omega}^r_R = (\Omega^r_{R/\mathbf{Z}})^\wedge_p$, the Sen sequence gives, for $n > 0$,

$$H^s(\mathrm{gr}^n_{\mathrm{N}} \mathbb{\Delta}_R) = \begin{cases} \widehat{\Omega}^{s-1}_R/(n-s+1), & 1 \le s < n, \\ \widehat{\Omega}^n_R, & s = n, \\ 0, & s \le 0 \text{ or } s > n. \end{cases}$$

Indeed, in degree $s - 1 < n$, $\Theta + n$ acts by the nonzero integer $n - s + 1$ on the p-torsion-free module $\widehat{\Omega}^{s-1}_R$; the top Hodge projection identifies the degree-n group.

Set $n = i + p$. After derived reduction modulo p, the only cohomology of G_{i+p} in degrees greater than i occurs in degrees $i + 1$ and $i + p$:

$$H^{i+1}(G_{i+p}) = \Omega^i_R/p, \qquad H^{i+p}(G_{i+p}) = \Omega^{i+p}_R/p.$$

Among $n - s + 1$ with $i < s < n$, the unique multiple of p is p, at $s = i + 1$; the extra Tor term produced by reducing the p-torsion belongs to degree i and does not occur in this range. The top group is the reduction of the p-torsion-free module $\widehat{\Omega}^{i+p}_R$.

Because $G_i \in D^{\le i}$, it does not change cofiber cohomology above i. The ring R is regular, so [4, Theorem 4.15] implies $Q_i \in D^{\ge i+1}$. The map q_i identifies the degree-$(i + p)$ group with $\mathscr{L}_{i+p}(R)$. Put $F_i(R) = \mathrm{fib}(q_i : Q_i(R) \to \mathscr{L}_{i+p}(R))$. It has a single cohomology group in degree $i + 1$, canonically identified with Ω^i_R/p; hence

$$F_i(R) \simeq \mathscr{L}_i(R)[-1]. \tag{1}$$

Here $H^{i+1}F_i$ is the cokernel of multiplication by p on $\widehat{\Omega}^i_R$, which fixes the normalization of (1).

Now animate the diagram of q_i, its nullhomotopy, its functorial fiber F_i, and (1). Modulo p, G_n preserves sifted colimits [2, Remark 5.5.10], as does $\mathscr{L}_n$ by the construction of derived exterior powers. Finite fibers and cofibers preserve sifted colimits in a stable category. Left Kan extension from polynomial rings therefore gives the coherent triangle for every animated ring. Rotating it defines β_i, together with the specified maps and nullhomotopies.

For polynomial compatibility, the decompositions over $R[X]$ are induced by the unit and exterior multiplication with dX. Multiplicativity identifies θ on each summand, and its Hodge-image nullhomotopy is compatible with dX. Thus there is a canonical decomposition

$$\begin{aligned} &(Q_i(R[X]) \xrightarrow{q_i} \mathscr{L}_{i+p}(R[X])) \\ &\quad \simeq (Q_i(R) \xrightarrow{q_i} \mathscr{L}_{i+p}(R))[X] \oplus (Q_{i-1}(R) \xrightarrow{q_{i-1}} \mathscr{L}_{i+p-1}(R))[X][-1]. \end{aligned}$$

Taking fibers gives the corresponding decomposition of $F_i(R[X])$. On the dX summand the Sen operator changes from Θ to $\Theta - 1$, giving the cokernel of multiplication by p on $\widehat{\Omega}^{i-1}_R dX$ in degree $i + 1$. Under (1), the map is exterior multiplication with dX, with coefficient one up to the fixed differential-ordering convention. After shifting the three fiber terms to degree zero, they lie in the heart of the derived module category; mapping spaces between such objects are discrete. The differential-form maps therefore lift uniquely to complexes with their commuting squares. Animation yields the stated decomposition of $\beta_i(R[X])$. □

3.2 The Frobenius–Witt splitting

We use Mao's Frobenius–Witt cotangent complex [17]. For a ring R, write FL_R for the pointed R/p-module of Frobenius–Witt differentials. There is a natural triangle

$$A_R \longrightarrow \mathrm{FL}_R \longrightarrow L_{R/\mathbf{Z}} \otimes^{\mathbf{L}}_{R,F} A_R \longrightarrow A_R[1]. \tag{2}$$

Mao proves that FL_V is a flat discrete V/p-module for every valuation ring V [17, Theorem A.2], and that $\mathrm{FL}_R = 0$ when the derived p-completion of R is perfectoid [17, Lemma A.1]. For a discrete ring the generators $w_R(x)$ obey

$$\begin{aligned} w_R(xy) &= x^p w_R(y) + y^p w_R(x), \\ w_R(x+y) &= w_R(x) + w_R(y) - P(x,y) w_R(p), \\ w_R(1) &= 0, \end{aligned}$$

where $P(x,y) = ((x+y)^p - x^p - y^p)/p$; see [17, Section 2 and Theorem 3.6].

Lemma 3.2. *Assume V is nondiscrete; put $A = V/p$ and $M = \mathrm{FL}_V$. Then $w_V(p) = 0$ and*

$$F_A^* C_V \simeq A[1] \oplus M.$$

Consequently, for every $n \geq 0$,

$$F_A^* \mathscr{L}_n(V) \simeq \bigoplus_{r=0}^{n} \bigwedge_A^r M[-r], \tag{3}$$

and every exterior power in this formula is flat.

Proof. Normalize $v(p) = 1$. Density gives $a \in V$ with $p^{-2} \leq v(a) < p^{-1}$ and $b = p/a^p$. Then a, b are in the maximal ideal and $a^{p^2} \in pV$. The Frobenius–Witt relations give

$$w_V(a^p) = p a^{p(p-1)} w_V(a) = 0, \qquad w_V(p) = w_V(a^p b) = a^{p^2} w_V(b) = 0.$$

Thus the first map in (2) vanishes. As M is discrete, the zero-point splitting is unique, giving $F_A^* C_V \simeq A[1] \oplus M$. Derived exterior powers and the usual décalage formula give (3). Flatness follows by writing M as a filtered colimit of finite free modules. □

3.3 The scalar calculation

After Frobenius pullback, β_i is a unit on each common exterior-power summand. Choose a, b as above and put

$$R_0 = (\mathbf{Z}_p[T,B]/(p - T^p B))_{(T,B)}, \qquad A_0 = R_0/p.$$

Evaluation $T \mapsto a$, $B \mapsto b$ gives a local map $R_0 \to V$. The ring R_0 is regular local of dimension two and p-torsion-free.

Lemma 3.3. *For every $j \geq 0$, the Frobenius pullback of $\beta_j(R_0)$, after scalar extension $A_0 \to A$, acts on the canonical H^0 copy of A by a unit $u_j \in A^\times$.*

Proof. By regularity and [17, Theorem 5.2], FL_{R_0} is flat over A_0. The Frobenius–Witt relation gives $w_{R_0}(p) = T^{p^2} w_{R_0}(B)$; this vanishes after base change to A because $a^{p^2} = 0$ in A. Thus the splitting of Lemma 3.2 yields a distinguished H^0 line, on which $F^*\beta_j$ acts by a scalar u_j.

Extend K to its completed algebraic closure, with valuation ring W. Then W is perfectoid and $A \to W/p$ is local. Mao's perfectoid vanishing yields $\mathrm{FL}_W = 0$, while Bhatt–Mathew's perfectoid Nygaard calculation gives $Q_j(W) = 0$ [4, Example 2.5]. Therefore $\beta_j(W)$ is an equivalence. Naturality shows that u_j becomes a unit in W/p; since $A \to W/p$ reflects units, u_j is already a unit. □

Lemma 3.4. *Under the splitting (3), the map*

$$H^r(F_A^*\beta_i(V)) : \bigwedge^r M \longrightarrow \bigwedge^r M, \qquad 0 \le r \le i,$$

is multiplication by the unit $\pm u_{i-r}$.

Proof. For $x_1, \dots, x_r \in V$, evaluate $R_0[X_1, \dots, X_r] \to V$, $X_s \mapsto x_s$. The polynomial compatibility in Lemma 3.1 identifies the pure r-differential summand of β_i with $\beta_{i-r}(R_0)[-r]$. Under Frobenius–Witt splitting, its generator maps to $w_V(x_1) \wedge \cdots \wedge w_V(x_r)$. Thus the map is $\pm u_{i-r}$ on these wedges, which generate $\bigwedge_V^r M$; the sign is a Koszul sign. □

3.4 Frobenius pullback reflects Tor bounds

Lemma 3.5. *Let $A = V/p$ for a rank-one mixed-characteristic valuation ring. Derived Frobenius pullback F_A^* reflects Tor-amplitude intervals and one-sided Tor lower bounds.*

Proof. Let W be the valuation ring of the completed algebraic closure of K, and put $D = W/p$. The map $A \to D$ is faithfully flat. Frobenius on D factors as

$$D \longrightarrow D/(\varpi) \xrightarrow{\sim} D, \qquad \varpi^p = p.$$

Base change along a nilpotent quotient reflects Tor bounds: when $J^N = 0$, each discrete module has a finite filtration with R/J-module subquotients. Hence F_D^* reflects Tor bounds. Frobenius naturality gives

$$(F_A^*E) \otimes_A^{\mathbf{L}} D \simeq F_D^*(E \otimes_A^{\mathbf{L}} D).$$

A bound on F_A^*E gives the same bound on $E \otimes_A^{\mathbf{L}} D$, which faithful flatness reflects to E. □

3.5 Weak F-smoothness

Theorem 3.6. *Let V be a p-complete rank-one valuation ring of mixed characteristic. Then V is p-quasisyntomic and weakly F-smooth in the sense of [4, Definition 4.2]. If V is nondiscrete, for every $i \ge 0$,*

$$F_A^*Q_i(V) \simeq \bigoplus_{r=i+1}^{i+p} \bigwedge_A^r M[-r], \qquad A = V/p, \quad M = \mathrm{FL}_V. \tag{4}$$

In particular $Q_i(V)$ has A-Tor amplitude bounded below by $i+1$. Moreover, for every twist $n \in \mathbf{Z}$ and $m \ge 0$,

$$\mathrm{fib}(\theta : \mathrm{gr}_{\mathrm{N}}^m \mathbb{\Delta}_V\{n\}/p \longrightarrow \mathrm{gr}_{\mathrm{N}}^{m+p} \mathbb{\Delta}_V\{n+p-1\}/p) \in D^{\ge m+2}, \tag{5}$$

$$\mathrm{fib}(\varphi_m : \mathrm{gr}_{\mathrm{N}}^m \mathbb{\Delta}_V\{n\}/p \longrightarrow \overline{\mathbb{\Delta}}_V\{m\}/p) \in D^{\ge m+2}. \tag{6}$$

Proof. First suppose V is nondiscrete. Apply F_A^* to Lemma 3.1. Lemma 3.2 splits $F_A^*\mathscr{L}_{i+p}(V) = L \oplus H$, where

$$L = \bigoplus_{r=0}^{i} \bigwedge^{r} M[-r], \qquad H = \bigoplus_{r=i+1}^{i+p} \bigwedge^{r} M[-r].$$

By Lemma 3.4, the restriction of $F_A^*\beta_i$ to L is an equivalence. A change of coordinates removes the off-diagonal map from H to the target, so the fiber is H. This proves (4). All exterior powers of M are flat, so the lower Tor bound is $i+1$; Lemma 3.5 reflects it to $Q_i(V)$.

The same argument applied to $F_A^*C_V = A[1] \oplus M$ shows that $L_{A/\mathbf{F}_p}$ has Tor amplitude $[-1, 0]$. Since V is p-complete and p-torsion-free, this is the cotangent condition for p-quasisyntomicity. The fiber of $\theta : G_i \to G_{i+p}$ is $Q_i[-1]$. Changing rings from A to V loses at most one Tor degree, since p is a nonzerodivisor, giving the weak F-smoothness bound of [4, Definition 4.2]. In particular $Q_i(V) \in D^{\geq i+1}$, so the untwisted θ-fiber lies in $D^{\geq i+2}$.

For twists, the associated graded Breuil–Kisin line has a canonical degree-zero unit, and multiplication identifies each twisted graded piece with the untwisted one; see [2, Corollary 5.5.14 and Remark 5.5.15]. The graded image of the syntomic Bott lift is θ times this unit [4, Construction 2.8 and Proposition 2.9]. This gives (5). Divided Frobenius is the telescope obtained by inverting θ on the Nygaard graded algebra [4, Proposition 2.6]. Successive cofibers have lower bounds tending to infinity, so the fiber from the first term to the telescope lies in $D^{\geq m+2}$, giving (6).

If V is discrete, it is a complete DVR and hence regular noetherian. The same assertions follow from the F-smoothness of regular rings [4, Theorem 4.15], along with the same graded-twist and telescope arguments. □

4 Bott strictness and Nygaard-completion defects

We isolate the discrepancy between Bott multiplication on the uncompleted and completed prismatic terms. It is measured by the twist-indexed defects E_k and by their truncated comparison with Bott cofibers in syntomic cohomology.

4.1 Finite Bott strictness on qrsp rings

Let R be a p-torsion-free quasiregular semiperfectoid (qrsp) ring. Put

$$D = \mathbb{\Delta}_R/p, \qquad N^r = \mathcal{N}^{\geq r}\mathbb{\Delta}_R/p, \qquad L = \mathbb{\Delta}_R\{1\}/p.$$

Locally choose a generator d of the prism ideal and a generator e of the Breuil–Kisin line, normalized by $\varphi_1(e) = (u/d)e$ with $u \in D^\times$. In this trivialization the mod-p Bott class is

$$b = ce^{p-1}, \qquad c = d/u,$$

up to a nonzero scalar in $\mathbf{F}_p$; see [4, Constructions 2.7–2.8, Proposition 2.9 and Remark 2.10].

Lemma 4.1 (Finite Bott strictness)**.** *Multiplication by c is injective on D and, for every $r \geq 0$,*

$$cD \cap N^{r+p} = cN^r. \tag{7}$$

Consequently, for every $q \geq p$ there is a short exact sequence

$$0 \longrightarrow D/N^{q-p} \xrightarrow{c} D/N^q \longrightarrow D/(cD + N^q) \longrightarrow 0. \tag{8}$$

Once Breuil–Kisin twists are restored, neither statement depends on the choices of d and e.

Proof. For a p-torsion-free qrsp ring, divided Frobenius identifies the Nygaard graded pieces with the relative conjugate filtration; these pieces inject into D after reduction modulo p, by [7, Theorem 12.2 and Proposition 12.8]. Moreover (p, d) is a regular sequence on the prismatic ring, so c is a nonzerodivisor modulo p. Since

$$\frac{\varphi(c)}{d^p} = u^{-p} \in D^\times,$$

we have $c \in N^p$, and multiplicativity of divided Frobenius shows that $c : \mathrm{gr}^r_{\mathrm{N}} D \to \mathrm{gr}^{r+p}_{\mathrm{N}} D$ is injective. If $cy \in N^{r+p}$, begin with $y \in N^0 = D$ and induct on $s < r$. If $y \in N^s$, the image of cy in $\mathrm{gr}^{s+p}_{\mathrm{N}} D$ vanishes, hence injectivity forces $y \in N^{s+1}$. Thus $y \in N^r$, proving (7). The quotient exact sequence follows. □

4.2 The completion defect

For a twist k, put

$$H_k(R) = \overline{\mathbb{\Delta}}_R\{k\}/p.$$

For $q \geq p$, let $B^k_q(R)$ be the cofiber of Bott multiplication

$$\frac{\mathbb{\Delta}_R\{k-p+1\}/p}{\mathcal{N}^{\geq q-p}} \xrightarrow{v_1} \frac{\mathbb{\Delta}_R\{k\}/p}{\mathcal{N}^{\geq q}}.$$

On a qrsp chart Lemma 4.1 identifies this with the ordinary quotient $D\{k\}/(ID\{k\} + N^q D\{k\})$. For a general quasisyntomic ring we take the cofiber as the definition; it is compatible with quasisyntomic descent and with filtered colimits modulo p. Define the induced filtration by

$$F^q H^0(H_k) = \ker(H^0 H_k \to H^0 B^k_q),$$

and set

$$\widehat{H}_k = \mathrm{R}\varprojlim_q B^k_q, \qquad E_k(R) = \mathrm{cofib}\big(H_k(R) \longrightarrow \widehat{H}_k(R)\big).$$

Lemma 4.2 (Bott fiber versus completion defect)**.** *Let $j \in \mathbf{Z}$ and $k = j + p - 1$. Define*

$$Y_j = \mathrm{fib}(b : \mathbb{\Delta}_R\{j\}/p \to \mathbb{\Delta}_R\{k\}/p),$$
$$\widehat{Y}_j = \mathrm{fib}(b : \widehat{\mathbb{\Delta}}_R\{j\}/p \to \widehat{\mathbb{\Delta}}_R\{k\}/p).$$

Then $Y_j \simeq H_k[-1]$ and $\widehat{Y}_j \simeq \widehat{H}_k[-1]$, whence

$$\mathrm{fib}(Y_j \to \widehat{Y}_j) \simeq E_k[-2]. \tag{9}$$

For p-torsion-free qrsp R, the objects H_k and $\widehat{H}_k$ are discrete, and E_k has amplitude in $[-1, 0]$.

Proof. On a qrsp chart, multiplication by b is multiplication by the nonzerodivisor c after trivializing the twist. Its cokernel is the Hodge–Tate quotient H_k, giving the first equivalence. Taking limits in (8) is exact since all transition maps in the quotient towers are surjective, and gives the second. Reduction modulo p commutes with these derived limits since $\mathbf{Z}/p$ is perfect over $\mathbf{Z}$. Quasisyntomic descent globalizes the identifications. The fiber of the map between the Bott fibers is (9). □

4.3 From graded bounds to completed Bott fibers

The weak F-smoothness bounds of Theorem 3.6 remain valid after Nygaard completion.

Lemma 4.3. *For $m \geq 0$ and every twist n,*

$$\mathrm{fib}(v_1 : \mathcal{N}^{\geq m}\widehat{\mathbb{\Delta}}_V\{n\}/p \to \mathcal{N}^{\geq m+p}\widehat{\mathbb{\Delta}}_V\{n+p-1\}/p) \in D^{\geq m+2}. \tag{10}$$

Moreover divided Frobenius factors naturally as

$$\mathcal{N}^{\geq n}\widehat{\mathbb{\Delta}}_V\{n\}/p \xrightarrow{\widetilde{\varphi}_n} H_n(V) \longrightarrow \widehat{H}_n(V),$$

with

$$\mathrm{fib}(\widetilde{\varphi}_n) \in D^{\geq n+2} \quad (n \geq 0). \tag{11}$$

Proof. The completed Nygaard filtration is complete by construction. The graded pieces of the fiber in (10) are the twisted θ fibers of (5), with indices at least m. Finite extensions and inverse limits preserve the common bound $m+2$, proving (10). For divided Frobenius, the associated graded map is the map to the θ-localized Hodge–Tate complex. Equation (6) gives the lower bound on every graded fiber. The filtered factorization through the completed source is that of [2, Remarks 5.7.8 and 5.7.10]. Passing to complete filtered objects gives (11). □

Put

$$C_j(V) = \mathrm{cofib}(v_1 : S_j(V) \to S_{j+p-1}(V)).$$

The following comparison is used in Sections 9 and 11.

Proposition 4.4 (Truncated Bott–defect comparison)**.** *Let $j \geq 1$ and $k = j+p-1$. Then*

$$\tau_{\leq j-1}C_j(V) \simeq \tau_{\leq j-1}(E_k(V)[-1]). \tag{12}$$

Equivalently, for $1 \leq q \leq j$, $H^{q-1}C_j(V) \simeq H^{q-2}E_k(V)$.

Proof. Syntomic cohomology is computed on Nygaard-completed terms by

$$S_i(V) \simeq \mathrm{fib}(\widehat{\varphi}_i - \mathrm{can} : \mathcal{N}^{\geq i}\widehat{\mathbb{\Delta}}_V\{i\}/p \to \widehat{\mathbb{\Delta}}_V\{i\}/p)$$

by [2, Proposition 7.4.6]. Taking Bott fibers between the two syntomic squares gives

$$F_j := \mathrm{fib}(S_j \to S_k) = \mathrm{fib}(\alpha - \beta : X_j \to \widehat{Y}_j),$$

where X_j is the Bott fiber on completed Nygaard sources. The divided-Frobenius contribution factors as $X_j \xrightarrow{f} Y_j \xrightarrow{g} \widehat{Y}_j$, $\alpha = gf$, and (11) implies $\mathrm{fib}(f) \in D^{\geq j+2}$. The canonical-map contribution β factors through

$$Z_j = \mathrm{fib}(v_1 : \mathcal{N}^{\geq j-1}\widehat{\mathbb{\Delta}}_V\{j\}/p \to \mathcal{N}^{\geq j+p-1}\widehat{\mathbb{\Delta}}_V\{k\}/p),$$

which belongs to $D^{\geq j+1}$ by (10). Applying $\tau_{\leq j}$, the map f becomes an equivalence, and the factorization through Z_j makes β zero in $D^{\leq j}$. Therefore $\tau_{\leq j}F_j \simeq \tau_{\leq j}\,\mathrm{fib}(g)$. By (9), $\mathrm{fib}(g) = E_k[-2]$; since $C_j = F_j[1]$, shifting gives (12). □

5 Separation on the first perfectoid overlap

In this section we prove that $E_k(V) \in D^{\geq 1}$ (Proposition 5.6). By descent this comes down to separatedness of the induced Nygaard filtration on the first nontrivial term of the perfectoid Čech nerve. We deduce it from an explicit bound for complete intersections over perfectoid rings, which depends only on the number of equations.

5.1 A separation estimate

Let $W = \mathcal{O}_C$ for an algebraically closed perfectoid field of mixed characteristic, and consider

$$S_r = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty}\rangle/(X_1, \dots, X_r).$$

Give $\overline{\mathbb{\Delta}}_{S_r}/p$ the relative conjugate filtration over W; the induced Nygaard filtration is the image of the prismatic Nygaard steps.

Lemma 5.1. *For every $s \geq 0$,*

$$\mathcal{N}^{\geq n}(\overline{\mathbb{\Delta}}_{S_r}/p) \cap \mathrm{Fil}_s^{\mathrm{conj}}(\overline{\mathbb{\Delta}}_{S_r}/p) = 0 \quad \text{if } n \geq ps + r(p-1) + p. \tag{13}$$

In particular, the induced Nygaard filtration is separated.

Proof. The qrsp calculation of [7, Lemmas 12.4 and 12.7] gives a basis indexed by $\mathbb{N}[1/p]^r$. In one variable the basis element e_u has Nygaard coefficient $t^{\max(n-\lfloor u\rfloor,0)}$ and relative conjugate degree $\lfloor u/p\rfloor$. Tensoring gives the multivariable basis. If the total conjugate degree is at most s, then

$$\sum_j \lfloor u_j\rfloor \leq p\sum_j \lfloor u_j/p\rfloor + r(p-1) \leq ps + r(p-1).$$

The displayed bound on n forces the Nygaard coefficient to be divisible by t^p. On Hodge–Tate specialization modulo p one has $t^p = 0$. This proves the intersection statement; exhaustiveness of the conjugate filtration gives separatedness. □

Lemma 5.2 (Separation for complete intersections)**.** *Let P be a p-torsion-free perfectoid W-algebra and $f_1, \dots, f_r \in P$ a sequence whose images are Koszul regular on P/p. Put $S = (P/(f_1, \dots, f_r))^\wedge_p$ and assume S is p-torsion-free. Then the image of $H^0(\mathcal{N}^{\geq n}\mathbb{\Delta}_S \otimes^{\mathbf{L}} \mathbf{F}_p)$ in $\overline{\mathbb{\Delta}}_S/p$ has zero intersection with $\mathrm{Fil}_s^{\mathrm{conj}}$ whenever $n \geq ps + r(p-1) + p$. The same conclusion holds for filtered colimits of such presentations with a uniform bound on r.*

Proof. First assume compatible p-power roots of the f_i have been chosen in P. Put

$$P_0 = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty}\rangle, \qquad S_0 = P_0/(X_1, \dots, X_r),$$

and map $P_0 \to P$ by sending X_i^{1/p^m} to the chosen roots of f_i. Koszul regularity gives a discrete derived base change

$$S_0/p \otimes^{\mathbf{L}}_{P_0/p} P/p \simeq S/p. \tag{14}$$

Use the perfectoid base-change statement in the proof of [7, Proposition 12.8]: prismatic cohomology and the relative conjugate filtration commute with derived p-complete base change along perfectoid algebras, and the transported good filtration is Nygaard. Thus, as filtered complexes,

$$\mathcal{N}^{\geq n}\mathbb{\Delta}_S \simeq \mathcal{N}^{\geq n}\mathbb{\Delta}_{S_0} \widehat{\otimes}^{\mathbf{L}}_{\mathbb{\Delta}_{S_0}} \mathbb{\Delta}_S. \tag{15}$$

Write every multi-index uniquely as $u = a + pk$, with $0 \le a_i < p$ and $k \in \mathbb{N}^r$; put $g_k = e_{pk}$. Multiplication by the finitely many e_a identifies the Hodge–Tate specialization as

$$\overline{\mathbb{\Delta}}_{S_0}/p = \bigoplus_{k\in\mathbb{N}^r} (S_0/p)g_k, \qquad \mathrm{Fil}_s^{\mathrm{conj}} = \bigoplus_{|k|\le s} (S_0/p)g_k. \tag{16}$$

The coefficient in $e_a g_k = u_{a,k} e_{a+pk}$ is a unit, since all new q-factorial factors have indices strictly between consecutive multiples of p. Hence (16) is a direct sum decomposition of S_0/p-modules. Write $H_{>s}$ for the complementary sum.

Some care is needed in the transport, since reducing a Nygaard step modulo the prism ideal can create negative cohomology; we therefore work with the whole morphism rather than with H^0. By [7, Lemma 12.7],

$$\mathcal{N}^{\ge n}\mathbb{\Delta}_{S_0} = \bigoplus_u t^{\max\left(n-\sum_i \lfloor u_i\rfloor, 0\right)} A_{\mathrm{inf}}(W) e_u,$$

where $t = \varphi^{-1}(\xi)$ and ξ generates the prism ideal of W. Each coefficient ideal is abstractly free of rank one, since t is a nonzerodivisor. As (p, ξ) is a regular sequence on $A_{\mathrm{inf}}(W)$, the Nygaard step is (p, ξ)-completely flat. Consequently

$$D_n := \mathcal{N}^{\ge n}\mathbb{\Delta}_{S_0} \otimes^{\mathbf{L}}_{A_{\mathrm{inf}}(W)} A_{\mathrm{inf}}(W)/(p, \xi) \tag{17}$$

is discrete. The estimate in the proof of Lemma 5.1 shows that the morphism of discrete P_0/p-modules $D_n \to \overline{\mathbb{\Delta}}_{S_0}/p$ factors through $H_{>s}$ whenever $n \ge ps + r(p-1) + p$. Tensor over P_0/p with P/p. By (14), the low and high sums in (16) remain discrete complementary free S/p-modules after derived base change. Equations (15) and (17) identify the transported source with the Nygaard map for S after Hodge–Tate reduction. Its source may now have negative cohomology, but the morphism still factors through the discrete high summand, so its image on H^0 meets the low summand trivially.

Remove the roots by André's flatness lemma [7, Theorem 7.14]. After a p-completely faithfully flat perfectoid extension $P \to P'$, all f_i acquire compatible roots. Regularity modulo p is preserved by this flat base change, and $S' = (P'/(f_1, \dots, f_r))^{\wedge}_p$ is a p-completely faithfully flat quasisyntomic extension of S. The conjugate filtration and its Hodge–Tate graded pieces base change along $S/p \to S'/p$, so $H^0(\overline{\mathbb{\Delta}}_S/p) \to H^0(\overline{\mathbb{\Delta}}_{S'}/p)$ is injective. Naturality gives the desired intersection statement for S.

Finally, for a filtered colimit of these complete intersections, finite Nygaard quotients and finite conjugate pieces commute with filtered colimits; both are built from finitely many Nygaard graded pieces, identified with finite conjugate pieces by divided Frobenius. A class and its witness of membership in a fixed Nygaard level occur at one finite stage, where the bound already holds. □

5.2 The first overlap

Let C be the completion of an algebraic closure of K, $W = \mathcal{O}_C$, and $R = (W \otimes_V W)^{\wedge}_p$. We prove separation of the induced Nygaard filtration on $\overline{\mathbb{\Delta}}_R/p$. We use Gabber–Ramero's description of tame and degree-p extensions [11, Proposition 6.2.12, Corollary 6.2.14 and Proposition 6.3.11]. After passage to the maximal tame extension, each finite Galois extension is built from a tower of degree-p extensions. At each stage the valuation ring is the filtered union of finite-free algebras given by one monic polynomial equation.

Lemma 5.3. *Let V^t be the valuation ring in the maximal tame extension K^t/K. Then*

$$B = (W \otimes_V V^t)^\wedge_p$$

is p-torsion-free perfectoid.

Proof. The tensor product is flat over W, hence p-torsion-free. The relative Frobenius

$$(V^t/p) \otimes^{\mathbf{L}}_{V/p,F} V/p \longrightarrow V^t/p$$

is an isomorphism. After strict henselization a finite tame cyclic extension of prime degree $\ell \neq p$ has a basis with distinct value cosets. Every nontrivial coset lattice is a directed union of principal lattices with values approaching zero; Frobenius carries this system isomorphically to that of the p-multiple coset. The filtered union is therefore relatively perfect. Base change to W/p identifies Frobenius $B/\pi \to B/p$, where $\pi = p^{1/p} \in W$. Since π is a nonzerodivisor, [5, Lemma 3.10] implies that B is perfectoid. □

Lemma 5.4 (Triangular finite models)**.** *Let $K_0 \subset K_1 \subset \cdots \subset K_r$ be a tower of henselian rank-one valued fields; suppose each K_i is tamely closed and $[K_i : K_{i-1}] = p$. Write $A_i = K_i^+$. Then A_r is a filtered union of A_0-subalgebras with triangular presentations*

$$A_0[X_1, \ldots, X_r]/(g_1, \ldots, g_r), \tag{18}$$

where $g_i \in A_0[X_1, \ldots, X_i]$ is monic of degree p in X_i. Each is finite free of rank p^r over A_0. After a p-torsion-free perfectoid base change $A_0 \to B$, the completed base changes are complete intersections with the same r equations, regular modulo p.

Proof. For a degree-p extension, the assertion is the filtered single-generator description of [11, Proposition 6.3.11]: when $K = K^t$ and $[E : K] = p$, the valuation ring E^+ is a filtered union of finite K^+-algebras $K^+[x]$ with integral primitive generator x and minimal polynomial monic of degree p.

Proceed inductively. For a finite subset $F \subset A_r$, choose an integral primitive element x_r for K_r/K_{r-1} such that $F \subset A_{r-1}[x_r]$. Only finitely many coefficients occur in the minimal polynomial and in the expressions of elements of F. By induction they lie in a single triangular algebra $D_{r-1} = A_0[x_1, \ldots, x_{r-1}] \subset A_{r-1}$. Its fraction field is K_{r-1}. The minimal polynomial $g_r \in D_{r-1}[X_r]$ is monic of degree p, and the map

$$D_r := D_{r-1}[X_r]/(g_r) \longrightarrow A_r$$

is injective: D_r is free of rank p over D_{r-1}, and its generic fiber is the field K_r. It contains F. Every finite subset thus lies in a model (18), and two such models embed in a third by applying the construction to the union of their generators. Successive monic bases $1, X_i, \ldots, X_i^{p-1}$ show that every model is finite free of rank p^r over A_0.

Choose polynomial representatives for coefficients of g_i in the preceding variables. Monicity makes g_i a nonzerodivisor after quotienting by earlier equations and after arbitrary base change. Hence $(g_1, \ldots, g_r)$ is a regular sequence, also modulo p when the resulting algebra is p-torsion-free. The completed perfectoid base changes remain finite free, so they are complete intersections with r equations.

In the application, intermediate fields remain tamely closed: for separable E/K, $E^t = EK^t$ [11, Section 6.2.18]; if $K = K^t$ and E/K occurs in a finite p-tower, this identity gives $E = E^t$. □

Proposition 5.5 (First-overlap separation)**.** *The induced Nygaard filtration on $H^0(\overline{\mathbb{\Delta}}_R/p)$ is separated.*

Proof. Let L/K range through finite subextensions of the algebraic closure and put $T_L = (W \otimes_V V_L)^\wedge_p$. Every Hodge–Tate class on R comes from one T_L, because $R/p = \operatorname{colim}_L T_L/p$ and the finite conjugate filtration commutes with this colimit.

After replacing L by a finite Galois extension, set $E = LK^t$. The extension E/K^t is a finite p-extension. Choose a tower

$$K^t = F_0 \subset F_1 \subset \cdots \subset F_r = E, \qquad [F_i : F_{i-1}] = p.$$

The intermediate fields remain tamely closed. Lemma 5.4 supplies a cofinal filtered system of subalgebras of E^+ with presentations $V^t[X_1, \dots, X_r]/(g_1, \dots, g_r)$, where g_i is monic of degree p in its new variable. After base change along $V^t \to B$ and p-completion these are p-torsion-free complete intersections, regular modulo p, with r equations; note that r depends only on L.

The map

$$T_L \longrightarrow (B \otimes_{V^t} E^+)^\wedge_p$$

is p-completely faithfully flat and quasisyntomic: valuation extensions are flat, and the cotangent complex has the Tor bound of [11, Theorem 6.5.8]. The target is a filtered colimit of these complete intersections. By Lemma 5.2, its Nygaard filtration is separated with a bound depending on the fixed tower length r and the conjugate degree. Quasisyntomic descent for finite quotients reflects separation to T_L.

To descend without changing the finite stage, let $T \to U$ be a p-completely faithfully flat quasisyntomic map. Both $\overline{\mathbb{\Delta}}_T/p$ and every finite quotient $B_q(T)$ are connective, by the Hodge–Tate filtration and the quasisyntomic Tor bound. Čech descent makes both maps $H^0(\overline{\mathbb{\Delta}}_T/p) \to H^0(\overline{\mathbb{\Delta}}_U/p)$ and $H^0B_q(T) \to H^0B_q(U)$ injective. Thus the induced filtration is strict on H^0:

$$H^0(\overline{\mathbb{\Delta}}_T/p) \cap F^qH^0(\overline{\mathbb{\Delta}}_U/p) = F^qH^0(\overline{\mathbb{\Delta}}_T/p). \tag{19}$$

Let $x \in H^0(\overline{\mathbb{\Delta}}_R/p)$ lie in every Nygaard level; choose L so that it comes from $x_L \in H^0(\overline{\mathbb{\Delta}}_{T_L}/p)$. The map $T_L \to R$ is a p-completely faithfully flat quasisyntomic valuation base change. By (19), x_L lies in every Nygaard level of T_L. Separation there gives $x_L = 0$ and hence $x = 0$. □

5.3 The initial defect bound

Let W be the perfectoid valuation cover of V and $A_a = (W^{\otimes_V (a+1)})^\wedge_p$ its Čech nerve. By Lemma 2.3 these rings are p-torsion-free qrsp, and the prismatic and syntomic functors we use satisfy descent along them.

Proposition 5.6 (Initial connectivity of the defects)**.** *If V is nondiscrete, then $E_k(V) \in D^{\geq 1}$ for every $k \in \mathbf{Z}$.*

Proof. On each qrsp term Lemma 4.2 gives $E_k(A_a) \in D^{[-1,0]}$. The ring $A_0 = W$ is perfectoid, so its induced filtration is complete and $E_k(A_0) = 0$. By Proposition 5.5, the induced filtration on A_1 is separated, hence $H^{-1}E_k(A_1) = 0$. Descent gives $E_k(V) \simeq \operatorname{Tot}(E_k(A_\bullet))$ and the spectral sequence

$$E_1^{a,b} = H^bE_k(A_a) \Longrightarrow H^{a+b}E_k(V).$$

Only rows $b = -1, 0$ occur. In total degree zero, the only possibilities $(0, 0)$ and $(1, -1)$ vanish by the observations above; in total degree -1 only $(0, -1)$ can occur and it also vanishes. Thus $E_k(V) \in D^{\geq 1}$. □

6 The Milnor endpoint and the fiber product

This section and the next prove injectivity in the endpoint degree (Theorem 7.5). Apart from the syntomic excision statement (Proposition 6.2), the argument is algebraic. Here we construct operations in degrees two and three; Section 7 extends them to all degrees.

6.1 Milnor and Dennis–Stein conventions

For a local ring R, let $K_*^M(R)$ be ordinary Milnor K-theory and $\widehat{K}_*^M(R)$ Kerz's improved theory. For each fixed m there exists M_m, independent of R, such that

$$|R/\mathfrak{m}_R| > M_m \quad \Longrightarrow \quad K_m^M(R) \xrightarrow{\sim} \widehat{K}_m^M(R); \tag{20}$$

see [15, Proposition 10(5)].

We use the absolute Dennis–Stein presentation of K_2. For a 3-fold stable commutative ring, $K_2(R)$ is generated by symbols $\langle a, b\rangle$, defined when $1 + ab \in R^\times$, with relations

$$\langle a, b\rangle = -\langle -b, -a\rangle, \tag{21}$$

$$\langle a, b\rangle + \langle a, c\rangle = \langle a, b + c + abc\rangle, \tag{22}$$

$$\langle a, bc\rangle = \langle ab, c\rangle + \langle ac, b\rangle. \tag{23}$$

This is van der Kallen's theorem [21, Theorem 7.1]. If R is local with more than five residue elements, it is 5-fold stable and the symbol map is an isomorphism

$$K_2^M(R) \xrightarrow{\sim} K_2(R), \tag{24}$$

by [21, Theorem 8.4 and Corollary 8.5]. When b is a unit, $\langle a, b\rangle$ corresponds to $\{1 + ab, b\}$; we call this the unit-entry formula. For units u, v,

$$\{u, v\} = -\{v, u\}, \qquad \{u, -u\} = 0, \qquad \{u, u\} = \{u, -1\}. \tag{25}$$

Conventions. We fix a finite range of degrees $0 \le n \le N$ with $N \ge 3$. Until the last part of the proof of Theorem 7.5, V is nondiscrete and

$$|k| > \max\{5, M_0, \dots, M_N\},$$

so that (20) and (24) apply to every local ring with residue field k that occurs below. All constructions are integral before the final reduction modulo p, and no division by 2 occurs. For $p = 2$ the class $\{-1\}$ need not vanish modulo p and is kept throughout; this is the role of e below.

Set

$$\begin{aligned} G &= K^\times/K^{\times p}, \qquad P = \ker(A_1 \to k^\times/k^{\times p}), \\ A_m &= K_m^M(V)/p, \qquad A = \bigoplus_{m\ge 0} A_m, \qquad J = PA \subset A, \end{aligned} \tag{26}$$

and let $e = \{-1\} \in A_1$. Every element of P is represented by a principal unit: if $\bar{u} = \bar{c}^p$ in k, then $u/c^p \in 1 + \mathfrak{m}$. By Lemma 2.2, each $g \in G$ has representatives of arbitrarily small positive valuation; a *sufficiently small representative* of g means such a representative of small enough valuation.

6.2 The fiber product and excision

Put $T = V \times_k V$ with projections $q_0, q_1 : T \to V$ and diagonal section $s : V \to T$. The ring T is local, with maximal ideal $\mathfrak{m} \times \mathfrak{m}$ and residue field k, so the conventions above apply to T.

Lemma 6.1 (Tor-unitality of the fiber product)**.** *The square*

$$\begin{array}{ccc} T & \xrightarrow{q_0} & V \\ {\scriptstyle q_1}\downarrow & & \downarrow \\ V & \longrightarrow & k \end{array}$$

is a Tor-unital Milnor square. Consequently the square of nonconnective algebraic K-theory spectra is homotopy cartesian, and

$$\ker\big(K_2(T) \xrightarrow{q_0} K_2(V)\big) \simeq K_2(V, \mathfrak{m}), \tag{27}$$

compatibly with q_1. Moreover q_0, q_1 and $T \to k$ are homological epimorphisms.

Proof. Nondiscreteness implies $\mathfrak{m}^2 = \mathfrak{m}$: for $0 \neq a \in \mathfrak{m}$ choose b with $0 < v(b) < v(a)$ and write $a = b(a/b)$. Every ideal of a valuation domain is a filtered union of principal ideals, hence $\mathfrak{m}$ is flat, and $0 \to \mathfrak{m} \to V \to k \to 0$ gives $\mathrm{Tor}_i^V(k, k) = 0$ for $i > 0$. Thus $k \otimes^{\mathbf{L}}_V k \simeq k$, that is, $\mathfrak{m}$ is Tor-unital.

Let $I = 0 \times \mathfrak{m} \subset T$. Then q_1 maps I isomorphically onto $\mathfrak{m}$, and $T/I \simeq V$ through q_0. Tor-unitality depends only on the ideal as a non-unital ring [18, Remark 1.9], so I is Tor-unital and $q_0 : T \to V$ is a homological epimorphism. Tamme's excision theorem [19, Theorem 2] makes the K-theory square cartesian. Since q_0 is split by s, taking π_2 gives (27). By symmetry q_1 is a homological epimorphism, and so is the composite $T \to V \to k$. □

The same square is excisive for syntomic cohomology; this is what gives Milnor excision in all degrees.

Proposition 6.2 (Syntomic excision on the fiber product)**.** *For every $n \in \mathbf{Z}$ and $r \geq 1$, the square*

$$\begin{array}{ccc} \mathbf{Z}/p^r(n)_{\mathrm{syn}}(T) & \longrightarrow & \mathbf{Z}/p^r(n)_{\mathrm{syn}}(V) \\ \downarrow & & \downarrow \\ \mathbf{Z}/p^r(n)_{\mathrm{syn}}(V) & \longrightarrow & \mathbf{Z}/p^r(n)_{\mathrm{syn}}(k) \end{array}$$

is homotopy cartesian.

Proof. Cotangent complexes. Let $T \to S$ be a homological epimorphism. Base change along $T \to S$ gives $L_{(S \otimes^{\mathbf{L}}_T S)/S} \simeq S \otimes^{\mathbf{L}}_T L_{S/T} \simeq L_{S/T}$, while $S \to S \otimes^{\mathbf{L}}_T S$ is an equivalence; hence $L_{S/T} = 0$. The transitivity triangle for $\mathbf{Z} \to T \to S$ then gives $L_{S/\mathbf{Z}} \simeq S \otimes^{\mathbf{L}}_T L_{T/\mathbf{Z}}$, and therefore $\bigwedge^i L_{S/\mathbf{Z}} \simeq S \otimes^{\mathbf{L}}_T \bigwedge^i L_{T/\mathbf{Z}}$ for all i. By Lemma 6.1 this applies to $S = V$ (via q_0 and via q_1) and to $S = k$. Tensoring the exact sequence $0 \to T \to V \oplus V \to k \to 0$ with $M = \bigwedge^i L_{T/\mathbf{Z}}$ gives a fiber sequence

$$M \longrightarrow (M \otimes^{\mathbf{L}}_T V) \oplus (M \otimes^{\mathbf{L}}_T V) \longrightarrow M \otimes^{\mathbf{L}}_T k,$$

so the square formed by $\bigwedge^i L_{-/\mathbf{Z}}$ on T, V, V, k is cartesian for every $i \geq 0$, and so is its p-completion.

Prismatic terms. The diffracted Hodge complex has a functorial increasing conjugate filtration with graded pieces $\big(\bigwedge^i L_{R/\mathbf{Z}}\big)^{\wedge}_p[-i]$, and the Nygaard graded pieces of the absolute prismatic complexes are, functorially in R, fibers of Sen operators on terms of this filtration; see [2, Construction

4.7.1 and Remark 5.5.8]. By induction on i, each step $\mathrm{Fil}_i^{\mathrm{conj}}$ gives a cartesian square, hence so does each Nygaard graded piece. Each finite Nygaard quotient is a finite iterated extension of graded pieces, so it gives a cartesian square, and so does the inverse limit of these quotients, the Nygaard-completed complex, together with its Nygaard steps.

Syntomic terms. Syntomic cohomology is the fiber of divided Frobenius minus the canonical map on these Nygaard-completed terms [2, Proposition 7.4.6]. Fibers preserve cartesian squares, and derived reduction modulo p^r is exact. □

Corollary 6.3 (Milnor excision)**.** *For every $n \geq 0$ and $r \geq 1$,*

$$\widehat{K}_n^M(T)/p^r \xrightarrow{\sim} \widehat{K}_n^M(V)/p^r \times_{\widehat{K}_n^M(k)/p^r} \widehat{K}_n^M(V)/p^r. \tag{28}$$

In the fixed degree range the same holds with ordinary Milnor K-theory, $\ker(A_n \to K_n^M(k)/p) = J_n$*, and*

$$q_1 : \ker(K_n^M(T)/p \xrightarrow{q_0} A_n) \xrightarrow{\sim} J_n. \tag{29}$$

Proof. Lüders–Morrow prove that for every local p-henselian ring R, the syntomic symbol is an isomorphism

$$\widehat{K}_n^M(R)/p^r \xrightarrow{\sim} H^n(\mathbf{Z}/p^r(n)_{\mathrm{syn}}(R)), \tag{30}$$

by [16, Theorem 3.1]. The rings V, k, and T are henselian: for T, a simple residue root lifts in each V-coordinate, and the lifts have the same residue. Moreover, $\mathbf{Z}/p^r(n)_{\mathrm{syn}}(k) \in D^{\geq n}$ by the logarithmic de Rham–Witt description in characteristic p [6, Proposition 8.4 and Corollary 8.21]. In particular H^{n-1} of the syntomic complex of k vanishes, so the long exact sequence of the cartesian square of Proposition 6.2 shows that H^n of the T-term is the fiber product of the other three. With (30) this is (28), and (20) replaces improved by ordinary Milnor K-groups.

For the kernel, let $\bar{A} = A/J$. Two lifts of an element of $k^\times$ to $V^\times$ differ by a principal unit, whose class lies in P; so $\bar{x} \mapsto \{x\}$ gives a well-defined additive map $k^\times \to \bar{A}_1$. It respects the Steinberg relations: if $\bar{a} \neq 0, 1$ and a lifts $\bar{a}$, then $1 - a$ is a unit lifting $1 - \bar{a}$, and $\{a, 1 - a\} = 0$ in A_2. Hence it extends to a ring map $K_*^M(k)/p \to \bar{A}$. This map and the reduction $\bar{A} \to K_*^M(k)/p$ are inverse to each other, since both rings are generated by symbols of units. Thus $J_n = \ker(A_n \to K_n^M(k)/p)$, and the kernel of the first projection of the fiber product (28) is $0 \times J_n$, proving (29). □

6.3 A universal degree-three class

The new input in degree three is the following class.

Lemma 6.4 (Universal three-variable class)**.** *Let R be a henselian local $\mathbf{Z}_p$-algebra, and let $x, y, z \in R$ with $1 + xyz \in R^\times$. There is a natural class*

$$D_R(x; y, z) \in \widehat{K}_3^M(R)/p,$$

equal to $\{1 + xyz, y, z\}$ when y, z are units. It is functorial in R and satisfies

$$D(x; y, z) = -D(x; z, y), \tag{31}$$

$$D(x; y, z_1 z_2) = D(xz_2; y, z_1) + D(xz_1; y, z_2), \tag{32}$$

$$D(x; y, zh^p) = D(xh^p; y, z), \tag{33}$$

$$D(x; y, u) = \langle xu, y\rangle\{u\} \quad (u \in R^\times), \tag{34}$$

$$D(x; y, y) = D(-xy; y, -1). \tag{35}$$

Identity (34) is asserted when the residue field of R has more than five elements, using (24).

Proof. On $X = \operatorname{Spec} \mathbf{Z}_p[X, Y, Z, (1 + XYZ)^{-1}]$ the rational Milnor class $\{1 + XYZ, Y, Z\}$ has zero residue at every codimension-one point. The only possible nonunit divisors are $Y = 0$ and $Z = 0$, where the first entry specializes to 1; the vertical divisor contributes no additional residue. The Gersten theorem for improved Milnor K-theory on smooth schemes over a complete DVR therefore gives a unique Nisnevich section extending the rational class [16, Theorem 4.1 and Corollary 4.4]. Pulling its henselian stalk to R defines D_R.

Specialization. The identities in this lemma, and later ones among D-classes, are all proved in the same way. Given an identity involving finitely many elements of R, take a polynomial ring over $\mathbf{Z}_p$ with one variable for each of them, together with auxiliary variables $s_1, \dots, s_6$, sent to elements of R with distinct residues. Invert the finitely many expressions that are required to be units, and let S be the henselization at the prime lying under the closed point of R. Then S is an essentially smooth henselian local $\mathbf{Z}_p$-algebra with a local map $S \to R$, and its residue field has more than five elements. The classes in question are defined on S and pull back to those on R. By Gersten injectivity [16, Corollary 4.4], an identity holds on S once it holds at the generic point of S, where all D-classes are ordinary Milnor symbols over a field. It then holds on R by functoriality.

At the generic point, (31) is skew symmetry and (32) is expansion of the last Milnor entry. Applying (32) repeatedly gives $D(X; y, h^j) = jD(Xh^{j-1}; y, h)$, so

$$D(x; y, zh^p) = D(xh^p; y, z) + D(xz; y, h^p) = D(xh^p; y, z) + pD(xzh^{p-1}; y, h),$$

which is (33). Identity (35) is $\{1+xy^2, y, y\} = \{1+xy^2, y, -1\}$, which follows from $\{y, y\} = \{y, -1\}$. For (34), the residue field of S has more than five elements, so by (24) both sides are classes in $\widehat{K}_3^M(S)/p$. At the generic point the unit-entry formula gives $\langle xu, y\rangle = \{1+xyu, y\}$, and multiplying by $\{u\}$ gives $D(x; y, u)$. □

6.4 Relative degree-two and degree-three operations

We first identify relative K_2 with the principal-unit ideal in A_2.

Lemma 6.5. *The natural map identifies*

$$K_2(V, \mathfrak{m})/p \xrightarrow{\sim} J_2 \subset A_2.$$

Proof. The coefficient sequence for relative K-theory reduces injectivity of $K_2(V, \mathfrak{m}; \mathbf{F}_p) \to K_2(V; \mathbf{F}_p)$ to surjectivity of $K_3(V; \mathbf{F}_p) \to K_3(k; \mathbf{F}_p)$. Geisser–Levine identify $K_3(k; \mathbf{F}_p)$ with the image of Milnor symbols [12, Theorem 8.1]; their entries lift to units of V, proving surjectivity. The universal-coefficient sequences contain natural injections of the integral groups modulo p into these coefficient groups, so $K_2(V, \mathfrak{m})/p \to K_2(V)/p$ is injective too.

By Matsumoto's theorem, $K_2(k) = K_2^M(k)$; every residue symbol lifts to V, so $K_2(V) \to K_2(k)$ is surjective. Reducing the relative exact sequence modulo p identifies the image of $K_2(V, \mathfrak{m})/p$ with $\ker(K_2(V)/p \to K_2(k)/p)$. By (24) and Corollary 6.3, this kernel is J_2. □

For $g \in G$ and $w = \{1 + a\} \in P$ a principal unit, choose a representative $\rho \in K^\times$ of g with $0 < v(\rho) < v(a)$ and put

$$d_g(w) = \langle a/\rho, \rho\rangle \in J_2. \tag{36}$$

Both entries lie in $\mathfrak{m}$, so the symbol maps to zero in $K_2(k)$, and it lies in J_2 by Lemma 6.5.

Lemma 6.6. *Formula (36) induces an $\mathbf{F}_p$-bilinear map $d : G \times P \to J_2$. If a unit $u \in V^\times$ represents a class of $A_1 \subset G$, then*

$$d_{\{u\}}(w) = w\{u\}. \tag{37}$$

Proof. Rewriting both terms with (21) and applying (22) gives, for a common sufficiently small ρ,

$$\left\langle \frac{a}{\rho}, \rho \right\rangle + \left\langle \frac{b}{\rho}, \rho \right\rangle = \left\langle \frac{a+b+ab}{\rho}, \rho \right\rangle,$$

so the construction is additive in the principal unit. If $1+a$ and $1+a'$ have the same class in A_1, then $1+a' = (1+a)c^p$ with $c \in V^\times$. Since k has characteristic p, $\bar{c}^p = 1$ forces $\bar{c} = 1$, so $c = 1+b$ is a principal unit, and additivity gives $d_g(1+a') = d_g(1+a) + p\, d_g(1+b) = d_g(1+a)$ in J_2.

Applying (23) repeatedly gives $\langle y, h^j\rangle = j\langle yh^{j-1}, h\rangle$ whenever the symbols are defined. If $\rho' = \rho h^p$ represents the same class in G, order the two so that h is integral; (23) and this identity give

$$\left\langle \frac{a}{\rho'}, \rho' \right\rangle = \left\langle \frac{a}{\rho}, \rho \right\rangle + p \left\langle \frac{a}{h}, h \right\rangle,$$

so $d_g(w)$ does not depend on ρ. If ρ, σ represent g, h with $v(\rho) + v(\sigma) < v(a)$, then (23) gives $\langle a/\rho\sigma, \rho\sigma\rangle = \langle a/\sigma, \sigma\rangle + \langle a/\rho, \rho\rangle$, which is additivity in g. Finally, for (37) choose $h \in K^\times$ with $0 < pv(h) < v(a)$ and take $\rho = uh^p$. Then (23) gives $\langle a/\rho, \rho\rangle = \langle a/u, u\rangle + p\langle a/h, h\rangle$, and the unit-entry formula gives $\langle a/u, u\rangle = \{1+a, u\} = w\{u\}$. □

For the degree-three operation, if $a \in \mathfrak{m}$ define

$$f_g(a, b) = D(a/\rho; b, \rho), \tag{38}$$

and if $b \in \mathfrak{m}$ define

$$f_g(a, b) = -D(-b/\rho; -a, \rho), \tag{39}$$

where ρ is a sufficiently small representative of g.

Lemma 6.7. *The two formulas agree if $a, b \in \mathfrak{m}$, are independent of ρ, and are additive in g.*

Proof. Write $a = \rho A$ and $b = \rho B$. At the generic point of the universal source, (38) is $\{1 + \rho^2 AB, \rho B, \rho\}$ and (39) is $-\{1 + \rho^2 AB, -\rho A, \rho\}$, so their difference is $\{1 + \rho^2 AB, -\rho^2 AB, \rho\} = 0$. By the specialization argument in the proof of Lemma 6.4, the two formulas agree. Independence under $\rho \mapsto \rho h^p$ follows from (33), and additivity in g from (32), choosing representatives whose product is sufficiently small. □

Proposition 6.8 (Degree-three relative operation)**.** *For every $g \in G$ there is an $\mathbf{F}_p$-linear map $\Theta_g : J_2 \to A_3$, additive in g, such that for $w \in P$, $a \in A_1$ and $g, h \in G$,*

$$\Theta_g(wa) = -d_g(w)a, \tag{40}$$

$$\Theta_h d_g(w) = -\Theta_g d_h(w), \tag{41}$$

$$\Theta_g d_g(w) = d_g(w)e. \tag{42}$$

Proof. Since T is local with residue field k and $|k| > 5$, it is 5-fold stable [21, Theorem 8.4], so $K_2(T)$ has the Dennis–Stein presentation (21)–(23). We first construct a homomorphism on $K_2(T)$.

Write a Dennis–Stein symbol as $r = (x, x + a)$ and $t = (y, y + b)$, with $a, b \in \mathfrak{m}$. Put $U = 1 + xy$ and $H = 1 + x(y + b)$; both are units when the endpoint symbols are defined. Set

$$\Phi_g\langle r, t\rangle = f_g\left(x, \frac{b}{U}\right) + f_g\left(\frac{a}{H}, y + b\right). \tag{43}$$

For a finite relation, take a single sufficiently small ρ dividing every coordinate difference that occurs, and write

$$r = (x, x + \rho A), \quad t = (y, y + \rho B), \quad u = (z, z + \rho C).$$

Formula (43) is then represented on the smooth polynomial localization in x, y, z, A, B, C, ρ obtained by inverting the endpoint and intermediate denominators. Each intermediate denominator has the same residue as its endpoint counterpart, so it specializes to a unit of V. The usual Dennis–Stein first- and second-variable expansions give

$$\langle x + a, y + b\rangle - \langle x, y\rangle = \left\langle x, \frac{b}{1 + xy}\right\rangle + \left\langle \frac{a}{1 + x(y + b)}, y + b\right\rangle,$$

and at the generic point each f_g-term of (43) is the corresponding symbol times $\{\rho\}$. So the generic value of (43) is

$$(\langle x + a, y + b\rangle - \langle x, y\rangle)\{\rho\}. \tag{44}$$

For (21), the generic image of $\langle r, t\rangle + \langle -t, -r\rangle$ is

$$(\langle x + \rho A, y + \rho B\rangle + \langle -y - \rho B, -x - \rho A\rangle - \langle x, y\rangle - \langle -y, -x\rangle)\{\rho\},$$

which vanishes by (21) in the two coordinates. For (22), let $w = t + u + rtu$. The first coordinate is $w_0 = y + z + xyz$ and the second is $w_0 + \rho W$, where

$$W = B + C + Ayz + xBz + xyC + \rho(ABz + AyC + xBC) + \rho^2 ABC.$$

The generic image of $\langle r, t\rangle + \langle r, u\rangle - \langle r, w\rangle$ is (22) for $(x + \rho A, y + \rho B, z + \rho C)$ minus (22) for (x, y, z), multiplied by $\{\rho\}$; hence it vanishes. For (23), the coordinate differences of tu, rt and ru are respectively

$$\rho(yC + zB + \rho BC), \quad \rho(xB + yA + \rho AB), \quad \rho(xC + zA + \rho AC).$$

The generic image of $\langle r, tu\rangle - \langle rt, u\rangle - \langle ru, t\rangle$ is the difference of the two instances of (23), multiplied by $\{\rho\}$, and is zero. By the specialization argument in the proof of Lemma 6.4, these equalities hold on V, including specializations with zero or nonunit entries; by Lemma 6.7 the choice of ρ does not matter. Thus (43) induces a homomorphism $\Phi_g : K_2(T) \to A_3$. It vanishes on $s_*K_2(V)$, since (43) is zero if $a = b = 0$.

Restrict to $\ker q_0$. Lemmas 6.1 and 6.5 identify this kernel modulo p with J_2 via q_1; the resulting map is Θ_g. For $w = \{1 + a\}$ and a unit u, the unit-entry formula gives $w\{u\} = \langle a/u, u\rangle$, so $\langle (0, a/u), (u, u)\rangle \in \ker q_0$ is the lift of $w\{u\}$. In (43) it has $x = 0$, $b = 0$ and $U = H = 1$, so

$$\Theta_g(w\{u\}) = f_g(a/u, u) = D(a/(u\rho); u, \rho) = -D(a/(u\rho); \rho, u) = -\langle a/\rho, \rho\rangle\{u\}$$

by (38), (31) and (34). This is (40). Next choose representatives ρ, σ of g, h with $v(\rho) + v(\sigma) < v(a)$. The lift of $d_g(w) = \langle a/\rho, \rho\rangle$ is $\langle (0, a/\rho), (\rho, \rho)\rangle$, and (43) and (38) give $\Theta_h d_g(w) = f_h(a/\rho, \rho) = D(a/(\rho\sigma); \rho, \sigma)$. Exchanging the roles of g and h and using (31) proves (41). If $g = h$, choose $2v(\rho) < v(a)$; then (35) and (34) give

$$\Theta_g d_g(w) = D(a/\rho^2; \rho, \rho) = D(-a/\rho; \rho, -1) = d_g(w)e. \qquad \square$$

7 All-degree Milnor operations and the endpoint

We extend the operations of Proposition 6.8 to all degrees. Section 7.1 is pure graded algebra; Section 7.2 applies it to V.

7.1 A graph normal form

Let $A = \bigoplus_{m\geq 0} A_m$ be a connected $\mathbf{F}_p$-graded-commutative algebra generated by A_1. Fix $e \in A_1$ such that

$$u^2 = ue \quad (u \in A_1), \tag{45}$$

with $e = 0$ for odd p. Let $P \subset A_1$, $J = AP$, and let $G \supset A_1$ be an $\mathbf{F}_p$-vector space. Choose a complement of A_1 in G with ordered basis $(t_i)_{i\in I}$, and let C be the graded-commutative A-algebra generated by the t_i subject to

$$t_i^2 = t_i e. \tag{46}$$

For a finite subset $S = \{i_1 < \cdots < i_r\}$ of I, put $t_S = t_{i_1} \cdots t_{i_r}$.

Lemma 7.1. *As a left or right A-module,*

$$C = \bigoplus_{S\subset I,\ |S|<\infty} At_S.$$

Consequently $JC = \bigoplus_S Jt_S$ and $A \cap JC = J$.

Proof. For odd p this is the squarefree basis of an exterior algebra. For $p = 2$, adjoining one generator is quotienting by the monic polynomial $T^2 - eT$, which gives a free module with basis $1, T$; this does not require e to be a nonzerodivisor. Adjoin finitely many variables successively and pass to the filtered union. □

Lemma 7.2 (Graph normal form)**.** *Suppose that for every i there are degree-one linear maps $R_i : J_m \to J_{m+1}$ $(m \geq 1)$ such that for homogeneous $q \in J$ and $a \in A$,*

$$R_i(qa) = (-1)^{\deg a} R_i(q)a, \tag{47}$$

$$R_j R_i(q) = -R_i R_j(q), \tag{48}$$

$$R_i^2(q) = R_i(q)e. \tag{49}$$

Put $d_i = R_i|_P$ and $B = C/(wt_i - d_i(w) : w \in P, i \in I)$. Then $A \to B$ is injective. For any graded vector-space splitting $A = J \oplus H$,

$$B \simeq A \oplus \bigoplus_{\varnothing\neq S\subset I} Ht_S \tag{50}$$

as graded vector spaces.

Proof. Give J its usual right A-module structure and let right multiplication by t_i act as R_i. Equations (47)–(49) are exactly the relations needed for this right action to descend from the free associative extension to C. This gives a right C-linear evaluation map

$$\epsilon : JC = J \otimes_A C \longrightarrow J, \qquad q \otimes c \longmapsto q \cdot c,$$

restricting to the identity on J. Its kernel is generated by the elements $qt_i - R_i(q)$, $q \in J$: these lie in $\ker \epsilon$, and for $S = \{i_1 < \cdots < i_r\}$ the element $qt_S - \epsilon(qt_S) = qt_S - R_{i_r} \cdots R_{i_1}(q)$ is the telescoping

sum of $(qt_{i_1} - R_{i_1}(q))t_{i_2}\cdots t_{i_r}$, $(R_{i_1}(q)t_{i_2} - R_{i_2}R_{i_1}(q))t_{i_3}\cdots t_{i_r}$, and so on. By Lemma 7.1 every element of JC is a sum of such terms qt_S.

It is enough to take $q \in P$: equation (47) with graded commutativity implies $R_i(aq) = aR_i(q)$, and J is generated by P as an A-ideal. Thus the ideal defining B is exactly $\ker\epsilon$. If $a \in A$ dies in B, then $a \in A \cap \ker\epsilon \subset A \cap JC = J$, but $\epsilon(a) = a$, hence $a = 0$. Splitting the coefficient of each nonconstant t_S into its J and H parts and evaluating the J part gives (50), including linear independence.

In degree n, only operations through degree $n-1$ are needed. One can truncate A above degree n and set the top-degree operations to zero; the omitted identities lie outside the range. □

7.2 Extending the degree-three operation

Lemma 7.3 (Twisted derivation extension)**.** *Let $M = T(E)/(Q)$ be a connected quadratic graded algebra with E in degree one. Let $f_0, f_1 : M \to A$ be graded algebra maps, and suppose $w_x = f_1(x) - f_0(x) \in P$ for $x \in E$. Suppose $d : P \to J_2$ and $\Theta : J_2 \to A_3$ satisfy*

$$\Theta(wa) = -d(w)a \quad (w \in P,\ a \in A_1). \tag{51}$$

Put $\delta(x) = d(w_x)$. Then

$$\begin{aligned} D(x_1\cdots x_m) = \sum_{i=1}^{m}(-1)^{m-i} & f_1(x_1)\cdots f_1(x_{i-1}) \\ & \cdot \delta(x_i)f_0(x_{i+1})\cdots f_0(x_m) \end{aligned} \tag{52}$$

defines $D : M_m \to J_{m+1}$. It satisfies

$$D(uv) = f_1(u)D(v) + (-1)^{\deg v}D(u)f_0(v). \tag{53}$$

If $s : A \to M$ is a section with $f_0 s = f_1 s = \mathrm{id}$, then

$$D(s(a)) = 0, \qquad D(zs(a)) = (-1)^{\deg a}D(z)a. \tag{54}$$

Proof. Formula (52) is multilinear on the tensor algebra. Separating the differentiated slot between words proves (53). For $x, y \in E$,

$$f_1(x)f_1(y) - f_0(x)f_0(y) = w_x f_0(y) + f_1(x)w_y \in J_2.$$

Using (51) and graded commutativity,

$$D(xy) = -\delta(x)f_0(y) + f_1(x)\delta(y) = \Theta\big(f_1(x)f_1(y) - f_0(x)f_0(y)\big).$$

If $r \in Q$, both $f_0(r)$ and $f_1(r)$ vanish, so $D(r) = 0$. The Leibniz rule gives $D(urv) = 0$ for arbitrary words, proving descent to M. Every summand contains a factor $\delta(x_i) \in J_2$, hence lands in J. On diagonal degree-one generators $w_{s(a)} = 0$, so (53) gives (54). □

Apply the lemma with

$$M = K_*^M(T)/p, \quad f_i = q_i, \quad s : K_*^M(V)/p \to K_*^M(T)/p, \quad (d, \Theta) = (d_g, \Theta_g).$$

Milnor K-theory modulo p has the required quadratic presentation by Steinberg relations, and (51) is (40). For a unit pair $t_i = (u_i, u_i') \in T^\times$, the operation is

$$\Phi_g^{(m)}\{t_1, \dots, t_m\} = \sum_{i=1}^{m} (-1)^{m-i} \{u_1', \dots, u_{i-1}'\} d_g\{u_i'/u_i\}\{u_{i+1}, \dots, u_m\}. \tag{55}$$

It vanishes on diagonal classes and satisfies the module rule (54).

To obtain operations on J itself, put $L_m = \ker(q_0 : K_m^M(T)/p \to A_m)$. By (29),

$$q_1 : L_m \xrightarrow{\sim} J_m \tag{56}$$

in the fixed degree range, so each $\alpha \in J_m$ has a unique lift $\widetilde{\alpha} \in L_m$. Define

$$R_g(\alpha) = \Phi_g^{(m)}(\widetilde{\alpha}) \in J_{m+1}; \tag{57}$$

the value lies in J_{m+1} because every summand of (55) contains a factor $d_g(w_i) \in J_2$.

Proposition 7.4 (All-degree operations)**.** *In the fixed degree range, the maps (57) are bilinear in g and satisfy*

$$R_g(w) = d_g(w) \quad (w \in P), \tag{58}$$
$$R_g(\alpha a) = (-1)^{\deg a} R_g(\alpha) a, \tag{59}$$
$$R_u(\alpha) = \alpha u \quad (u \in A_1), \tag{60}$$
$$R_h R_g(\alpha) = -R_g R_h(\alpha), \tag{61}$$
$$R_g^2(\alpha) = R_g(\alpha) e. \tag{62}$$

Proof. Bilinearity in g follows from (55) and Lemma 6.6. For $w = \{1 + a\} \in P$, the lift of w is the class of $(1, 1 + a) \in T^\times$, and (55) in degree one gives (58). If $\widetilde{\alpha}$ lifts α, then $\widetilde{\alpha} s_*(a)$ lifts αa, so (54) gives (59).

Every element of J_m is a sum of terms wa with $w \in P$ and $a \in A_{m-1}$. For $u \in A_1 \subset G$, (37) and (59) give

$$R_u(wa) = (-1)^{m-1} d_u(w) a = (-1)^{m-1} wua = wau,$$

by graded commutativity. This proves (60). For $wa \in J_2$, (59) reads $R_g(wa) = -d_g(w)a$, which is (40); since these products span J_2, $R_g|_{J_2} = \Theta_g$, and (41), (42) become

$$R_h d_g(w) = -R_g d_h(w), \qquad R_g d_g(w) = d_g(w) e. \tag{63}$$

For homogeneous $a \in A$, applying (59) twice to a generator $wa \in J$ gives

$$R_h R_g(wa) = R_h d_g(w)\, a, \qquad R_g R_h(wa) = R_g d_h(w)\, a,$$

with the same total Koszul sign; so (63) proves (61). Similarly $R_g^2(wa) = R_g d_g(w)\, a = d_g(w) ea = R_g(wa) e$, which proves (62). □

7.3 Endpoint injectivity

Theorem 7.5 (Milnor endpoint)**.** *Let V be a p-complete henselian rank-one valuation ring of mixed characteristic $(0,p)$. For every $n \geq 1$,*

$$\widehat{K}_n^M(V)/p \hookrightarrow K_n^M(\operatorname{Frac} V)/p \tag{64}$$

is injective. Under the syntomic symbol and the norm-residue isomorphism, this is injectivity of

$$H^n S_n(V) \longrightarrow H^n E_n(V). \tag{65}$$

Proof. First assume that V is nondiscrete and that k satisfies the conventions of Section 6.1 for some $N \geq \max\{3, n\}$; put $A = K_*^M(V)/p$ in this range. The map $A_1 \to G$ is injective: if a unit is a pth power in K, its pth root has valuation zero and is a unit. Choose a complement $G = A_1 \oplus \bigoplus_i \mathbf{F}_p t_i$.

By (59), (61) and (62), the maps R_{t_i} satisfy the hypotheses of Lemma 7.2, and $u^2 = ue$ in A by (25). So there is a graded algebra B containing A injectively and generated over A by degree-one classes t_i, with relations $wt_i = d_{t_i}(w)$ and $t_i^2 = t_i e$. Every $g \in G$ then satisfies $wg = d_g(w)$ and $g^2 = ge$ in B. Indeed, write $g = u + \sum_i c_i t_i$ with $u \in A_1$ and $c_i \in \mathbf{F}_p$. The first relation follows from (37) and linearity of d in g. For the second, degree-one elements anticommute, so $g^2 = u^2 + \sum_i c_i^2 t_i^2 = (u + \sum_i c_i^2 t_i)e$, which is ge because $e = 0$ for odd p and $c_i^2 = c_i$ for $p = 2$.

We verify the field Steinberg relations $[x][1-x] = 0$ in B_2. If $v(x) > 0$, choose ρ representing $[x] \in G$ with $0 < v(\rho) < v(x)$, and write $x = \rho h^p$, so $v(h) > 0$. Since $1 - x$ is a principal unit, the relation $wg = d_g(w)$ and the Dennis–Stein identities (21), (23) give

$$[1-x][x] = d_{[x]}(1-x) = \langle -h^p, \rho\rangle = -\langle -\rho, h^p\rangle = -p\langle -\rho h^{p-1}, h\rangle = 0,$$

and $[x][1-x] = -[1-x][x] = 0$. If $v(x) = 0$ and $v(1-x) > 0$, apply this case to $1 - x$. If x and $1 - x$ are both units, the relation holds in A_2. If $v(x) < 0$, put $z = x^{-1}$. Then $1 - x = -x(1-z)$, so $[1-x] = e + [x] + [1-z]$ in G. Since $v(z) > 0$, the first case gives $[x][1-z] = -[z][1-z] = 0$, hence

$$[x][1-x] = [x]e + [x]^2 = 2[x]e = 0,$$

because $e = 0$ for odd p, while $2 = 0$ for $p = 2$. Thus the Milnor presentation of the field defines a graded map $K_*^M(K)/p \to B$. Its composite with $A_n \to K_n^M(K)/p$ is the injection $A_n \hookrightarrow B_n$, since both send $\{u\}$ to u. By (20) this proves (64) for large residue fields.

If k is finite, only finitely many residue-cardinality bounds are needed through degree N. Choose a finite extension k'/k of degree d prime to p and large enough to meet them. Henselianity lifts it to a finite étale local valuation extension V'/V of the same degree. The value group is unchanged, so the nondiscrete argument applies to V'. If $x \in H^n S_n(V)$ dies generically, its restriction to V' dies generically and is zero by the large-residue case. The finite étale syntomic norm of [16, Lemma 3.3 and Corollary 3.4] satisfies

$$\operatorname{Nm}_{V'/V} \circ \operatorname{res}_{V'/V} = d$$

on syntomic cohomology. Since d is invertible modulo p, restriction is injective and $x = 0$.

If V is discrete, it is a complete discrete valuation ring (its p-adic and $\mathfrak{m}$-adic topologies agree), hence regular, and the endpoint is covered by the comparison theorem for regular rings [4, Theorem 4.15 and Proposition 5.2].

Finally, (30) identifies the source of (64) with $H^n S_n(V)$, and the norm-residue isomorphism identifies $K_n^M(K)/p$ with $H^n_{\acute{e}t}(K, \mu_p^{\otimes n})$ [22, Theorem 6.16]. Under these identifications (64) becomes (65): both composites $K_n^M(V)/p \to H^n_{\acute{e}t}(K, \mu_p^{\otimes n})$ are multiplicative, and in degree one both send a unit to its Kummer class. □

8 Frobenius base change and twist periodicity

It remains to control the completion defects E_k. In this section we show that after the base change ψ^* below, which is conservative, the defects for different twists become equivalent. The proof uses the Hodge–Tate stack of V. Put

$$A = V/p, \qquad \psi : V \longrightarrow A, \quad v \longmapsto \bar{v}^p, \qquad \psi^*(-) = - \otimes^{\mathbf{L}}_{V,\psi} A.$$

Throughout the section V is nondiscrete, complete and rank one.

8.1 Finite-Tor base change

Lemma 8.1 (Base change for convergent inverse systems)**.** *Let B have Tor amplitude $[-d, 0]$ over R, and let (X_i) be an inverse system such that for each i the fibers $\mathrm{fib}(X_j \to X_i)$ for $j \geq i$ lie in $D^{\geq c_i}$, where $c_i \to \infty$. Then*

$$B \otimes^{\mathbf{L}}_R \mathrm{R}\varprojlim_i X_i \xrightarrow{\sim} \mathrm{R}\varprojlim_i (B \otimes^{\mathbf{L}}_R X_i).$$

The same holds for the totalization of a cosimplicial object whose terms are uniformly bounded below.

Proof. Put $\widehat{X} = \mathrm{R}\varprojlim_i X_i$. Taking the inverse limit in the fiber triangles over $j \geq i$ yields $\mathrm{fib}(\widehat{X} \to X_i) \in D^{\geq c_i}$. Tensoring with B lowers connectivity by at most d; the same estimate applies to the inverse limit of the base-changed tower. Hence the canonical map is an isomorphism in any fixed cohomological degree for $i \gg 0$. For totalizations, normalized cosimplicial degree r is a summand of the rth term and appears with a shift $[-r]$. Thus successive partial-totalization fibers become arbitrarily connective, and the inverse-system argument applies. □

Lemma 8.2 (Conservativity of ψ^*)**.** *The Frobenius pullback $F_A^* : D(A) \to D(A)$ is conservative; hence $\psi^* : D(V) \to D(A)$ is conservative on derived p-complete complexes. The V-module A via ψ has Tor amplitude $[-1, 0]$. Moreover, if a V-module M is annihilated by a power of p and*

$$\mathrm{Tor}_1^V(M, A; \psi) = 0,$$

then $M = 0$.

Proof. A valuation ring has weak dimension at most one, giving the Tor-amplitude assertion. Extend the valuation to $K(p^{1/p})$ and let V' be its valuation ring. It is faithfully flat over V. Take $\pi \in V'$ with $\pi^p = p$, and put $A' = V'/p$. Frobenius on A' factors as

$$A' \longrightarrow V'/\pi \longrightarrow A'.$$

The first arrow has nilpotent kernel. The second, induced by $x \mapsto x^p$, is faithfully flat: V'/π is a quotient of a valuation ring, its finitely generated ideals are principal, and the annihilator criterion

applies as in [17, Proposition A.4 and Corollary A.5]; see also [13]. Derived base change by a nilpotent quotient is conservative, so $F^*_{A'}$ is conservative; faithfully flat descent along $A \to A'$ proves the assertion for F^*_A.

For derived p-complete N, $\psi^* N \simeq F^*_A(N \otimes^{\mathbf{L}}_V A)$. Thus $\psi^* N = 0$ implies $N/p = 0$, hence $N = 0$ by derived p-completeness.

For the last statement, base change to V' and factor ψ' through $B = V'/\pi$ and then the faithfully flat map $B \to A'$. The two-term resolution of B over V' gives

$$\mathrm{Tor}_1^{V'}(M \otimes_V V', A'; \psi') \simeq (M \otimes_V V')[\pi] \otimes_B A'.$$

Its vanishing implies that multiplication by π is injective on $M \otimes_V V'$. But $p^N M = 0$ for some N, so π^{pN} annihilates the base change. An injective nilpotent endomorphism forces it to vanish, and faithful flatness gives $M = 0$. □

8.2 Twisted Nygaard filtrations

We use the multiplicative induced twisted Nygaard filtration of [4, Constructions 3.1 and 3.3 and Remark 3.4] as a filtration on complexes; its graded pieces alone would not suffice. For every $k \in \mathbf{Z}$,

$$F^0 H_k = H_k, \qquad F^r H_k \otimes F^s H_\ell \longrightarrow F^{r+s} H_{k+\ell}. \tag{66}$$

All-integer twists and multiplicativity follow from filtered Breuil–Kisin twists [2, Section 5.5]. On qrsp charts this filtration is induced by images of Nygaard ideals in Hodge–Tate cohomology, with finite quotients B^k_q as in Section 4.

Lemma 8.3 (Completion commutes with ψ^*)**.** *For every k,*

$$\psi^* \widehat{H}_k \xrightarrow{\sim} \mathrm{R}\varprojlim_q \psi^* B^k_q.$$

Consequently $\psi^* E_k$ is the completion defect of the base-changed filtered object $\psi^* H_k$.

Proof. By Theorem 3.6, the twisted Bott graded fibers become increasingly connective. The induced filtration satisfies $\mathrm{gr}^j_F H_k \in D^{\geq j-p+1}$ for $j \geq p$, uniformly in k; this is the form of weak F-smoothness used in [4, Remark 4.5]. Thus for $q' \geq q \geq p$,

$$\mathrm{fib}(B^k_{q'} \to B^k_q) \in D^{\geq q-p+1}.$$

The V-module A via ψ has Tor amplitude $[-1, 0]$ by Lemma 8.2. Apply Lemma 8.1; exactness of ψ^* identifies the completion cofibers. □

If a twist-one class becomes a unit after ψ base change, (66) makes it a filtered unit.

Lemma 8.4 (Filtered unit)**.** *Suppose $\eta \in H^0(\psi^* H_1)$ and $\eta^{-1} \in H^0(\psi^* H_{-1})$ multiply to 1. Multiplication by η gives inverse equivalences*

$$\psi^*(F^\bullet H_k) \xrightarrow{\sim} \psi^*(F^\bullet H_{k+1})$$

for every k, and equivalences of their completion defects.

Proof. Both classes are in filtration zero because $F^0 H_{\pm 1} = H_{\pm 1}$. Multiplicativity gives maps on every filtration step, whose products are the identity in $H^0(F^0 H_0)$. Thus the two filtered multipliers are inverses in the filtered derived category. Finite quotients, inverse limits, and cofibers preserve equivalences, and Lemma 8.3 identifies the defects with $\psi^* E_k$. □

8.3 A geometric trivialization of the Breuil–Kisin line

We construct the required unit. The quotient presentation of Hodge–Tate prismatization for a regular local prism in [3, Construction 9.4 and Proposition 9.5] is the geometric input. For a bounded prism $(P,(E))$ presenting a complete regular local ring, the Hodge–Tate Cartier–Witt stack is a quotient of a $G_a^\sharp$-torsor by the semidirect product of the divided-power tangent group and $G_m^\sharp$, and the $G_m^\sharp$ character on the Breuil–Kisin line is the zeroth Witt coordinate; compare [2, Section 4.4].

Lemma 8.5. *There exist $a,b\in\mathfrak{m}_V$ with*

$$a^p b = p, \qquad a^{p^2}\in pV.$$

For

$$P=\mathbf{Z}_p[[T,B]], \qquad E=p-T^pB, \qquad R=P/(E),$$

with $\delta(T)=\delta(B)=0$, the pair $(P,(E))$ is a bounded prism and R is a p-torsion-free complete regular local ring of dimension two. Evaluation $T\mapsto a$, $B\mapsto b$ defines a continuous map $R\to V$.

Proof. Normalize $v(p)=1$. By density choose $p^{-2}\le v(a)<p^{-1}$ and put $b=p/a^p$. Then $a,b\in\mathfrak{m}_V$, $a^pb=p$ and $a^{p^2}\in pV$. The ring P is regular local of dimension three, and $E\in(p,T,B)\setminus(p,T,B)^2$, so R is regular local of dimension two. Modulo p, $E=-T^pB$ is a nonzerodivisor in $\mathbf{F}_p[[T,B]]$, proving p-torsion-freeness of R. Finally $\delta(E)$ has unit constant term $\delta(p)=1-p^{p-1}$; hence $p\in(E,\varphi(E))$. Completeness and boundedness are immediate, and the evaluation converges because $v(a),v(b)>0$. □

Lemma 8.6 (Hodge–Tate global sections after ψ base change)**.** *Let $X=\operatorname{Spf}V$ and X^{HT} be its absolute Hodge–Tate stack. Put*

$$Z=\operatorname{Spec}A\times_{X,\psi}X^{\mathrm{HT}},$$

with derived fiber product. For every $k\in\mathbf{Z}$, the sections of the pulled kth Breuil–Kisin line identify multiplicatively with

$$R\Gamma(Z,\mathcal{O}\{k\})\simeq\psi^*\overline{\mathbb{\Delta}}_V\{k\}. \tag{67}$$

Consequently, a trivialization of the line and its dual on Z gives inverse classes after canonical derived reduction $\psi^*\overline{\mathbb{\Delta}}_V\{\pm1\}\to\psi^*H_{\pm1}$.

Proof. Choose the perfectoid valuation cover $V\to W$ of Lemma 2.3 and set $R_a=(W^{\otimes_V(a+1)})_p^\wedge$. These are p-torsion-free qrsp rings. By prismatization comparison and flat-local descent [3, Lemma 6.3, Theorem 7.17, Corollary 7.18 and Corollary 8.17], X^{HT} is covered by derived formal affines with rings $\overline{\mathbb{\Delta}}_{R_a}$, whose cosimplicial object is the full Čech nerve. The kth Breuil–Kisin line has a discrete module of sections on each qrsp chart. These section complexes have a common lower bound, and descent identifies the totalization with $\overline{\mathbb{\Delta}}_V\{k\}$.

After derived base change by ψ, the charts remain derived affine and the local sections become $\psi^*\overline{\mathbb{\Delta}}_{R_a}\{k\}$. It remains to commute ψ^* with the infinite totalization. Normalized cosimplicial degree a is a summand of the ath term and appears with shift $[-a]$; as the unbased terms are uniformly connective, partial-totalization fibers become arbitrarily connective. Since ψ^* has Tor amplitude $[-1,0]$ by Lemma 8.2, the totalization case of Lemma 8.1 applies. Flat-local descent gives (67). The identifications come from twist-graded multiplication and are multiplicative in k.

Note that (67) concerns the integral Hodge–Tate line, whereas

$$\psi^* H_k = \psi^*(\overline{\mathbb{\Delta}}_V\{k\}/p) \simeq \psi^* \overline{\mathbb{\Delta}}_V\{k\} \otimes^{\mathbf{L}}_{\mathbf{Z}_p} \mathbf{F}_p.$$

Although A has characteristic p, this derived reduction can have a Tor term, so the two are different. But the reduction map is multiplicative, so inverse sections of the integral lines remain inverse on $\psi^* H_{\pm 1}$. □

Proposition 8.7 (Frobenius-pulled Breuil–Kisin-line unit)**.** *There is a unit $\eta \in H^0(\psi^* H_1)$ with inverse in $H^0(\psi^* H_{-1})$. Consequently,*

$$\psi^* E_k \simeq \psi^* E_\ell \qquad (k, \ell \in \mathbf{Z}). \tag{68}$$

Proof. Let $f : \operatorname{Spec} A \to \operatorname{Spf} R$ be given by $T \mapsto a^p \bmod p$, $B \mapsto b^p \bmod p$. Consider the base change of the Hodge–Tate Cartier–Witt stack of R along f. Its quotient presentation is an equivalence over $\operatorname{Spf} R$, and the torsor and the acting divided-power groups are flat over the base, so termwise base change computes the derived base change. For an A-algebra S, the adjoint δ-map $P \to W(S)$ sends

$$T \longmapsto [a^p], \qquad B \longmapsto [b^p].$$

Since $a^{p^2} = 0$ in A, it sends E to p. The pulled Cartier–Witt divisor is the standard divisor $p : W(S) \to W(S)$. The torsor of splittings in [3, Proposition 9.5] is therefore

$$\{\tau \in W(S) : p\tau = p\} = 1 + W(S)[F] = G_m^\sharp(S).$$

The divided-power tangent action is controlled by

$$dE = -pT^{p-1} B\, dT - T^p\, dB.$$

Both coefficients vanish after applying f: the first because $p = 0$, the second because $a^{p^2} = 0$ in A. Thus the tangent group acts trivially. If y is the zeroth Witt coordinate of τ, it is invertible and $y(u\tau) = u_0 y(\tau)$ for $u \in G_m^\sharp$. Hence y is a nowhere-vanishing equivariant section of the character defining the Breuil–Kisin line (or its dual, according to sign convention), trivializing it after f.

Pull the trivialization along $R \to V$ to $Z = \operatorname{Spec} A \times_{\operatorname{Spf} V, \psi} (\operatorname{Spf} V)^{\mathrm{HT}}$. By Lemma 8.6, its twist-graded sections are the derived ψ-pullbacks of the integral Hodge–Tate Breuil–Kisin lines, compatibly with multiplication. Thus there are inverse sections in $H^0\psi^*(\overline{\mathbb{\Delta}}_V\{1\})$ and $H^0\psi^*(\overline{\mathbb{\Delta}}_V\{-1\})$. By the last part of Lemma 8.6 they give inverse classes in $\psi^* H_{\pm 1}$. Now apply Lemmas 8.3 and 8.4. Multiplication by η identifies adjacent filtered twists and their defects; iteration gives (68). □

9 Vanishing of the Nygaard-completion defects

We now combine the endpoint theorem (Theorem 7.5) with twist periodicity (Proposition 8.7).

Theorem 9.1 (Defect vanishing)**.** *Let V be a nondiscrete complete rank-one valuation ring of mixed characteristic. Then $E_k(V) = 0$ for every $k \in \mathbf{Z}$.*

Proof. Proposition 5.6 gives $E_k \in D^{\geq 1}$ for all k. We prove inductively

$$E_k \in D^{\geq s} \qquad \text{for all } k \tag{69}$$

for every $s \geq 1$. Assume (69). Fix a weight $j \geq s+2$. By Proposition 4.4, $H^s C_j \simeq H^{s-1} E_{j+p-1} = 0$. Thus the Bott map

$$H^{s+1} S_j(V) \longrightarrow H^{s+1} S_{j+p-1}(V) \tag{70}$$

is injective. The same argument applies to every weight $j' \geq j$, so all the maps in the Bott tower starting in weight j are injective on H^{s+1}. Bhatt–Mathew's Bott localization theorem identifies the telescope with generic étale cohomology:

$$\operatorname*{colim}_{m \geq 0} S_{j+m(p-1)}(V) \xrightarrow{\sim} E_j(V), \tag{71}$$

with the usual coefficient identifications [4, Theorem 5.1]. Since all transition maps on H^{s+1} are injective, the comparison $H^{s+1} S_j(V) \to H^{s+1} E_j(V)$ is injective. It is surjective by Proposition 2.6, since $s+1 < j$. Thus it is an isomorphism for every $j \geq s+2$.

Take $j = s+2$ and $k = j+p-1$. The Bott map on H^{s+1} is an isomorphism because both comparison maps are isomorphisms in that degree and the generic Bott class is invertible. On H^{s+2}, Theorem 7.5 makes the weight-$(s+2)$ comparison injective. Naturality with the generic Bott class makes $H^{s+2} S_j(V) \to H^{s+2} S_k(V)$ injective. The long exact cohomology sequence of C_j gives $H^{s+1} C_j = 0$. Apply Proposition 4.4 with $q = j = s+2$:

$$H^s E_{s+p+1} = 0. \tag{72}$$

This defect belongs to $D^{\geq s+1}$.

Pull back by ψ. Since its Tor amplitude is $[-1, 0]$, (72) implies $\psi^* E_{s+p+1} \in D^{\geq s}$. By (68), $\psi^* E_k$ has this same bound for every k. From the induction hypothesis and the one-degree Tor spectral sequence,

$$H^{s-1}(\psi^* E_k) \simeq \operatorname{Tor}_1^V(H^s E_k, A; \psi) = 0.$$

The module $H^s E_k$ is annihilated by p. Lemma 8.2 forces $H^s E_k = 0$. Therefore $E_k \in D^{\geq s+1}$ for every k, closing the induction. Each E_k is arbitrarily connective and thus zero. □

Remark 9.2. The proof of Theorem 7.5 uses neither weak F-smoothness nor the defects E_k, and Sections 3–5 do not use the endpoint theorem. The two meet only at (72).

10 F-smoothness of valuation rings

We now extend the rank-one results to valuation rings of arbitrary rank by excision. Weak F-smoothness and Nygaard completeness are treated separately: Theorem 3.6 provides the former in rank one, and Theorem 9.1 the latter. For the Tor-amplitude condition we also need a localization argument (Lemma 10.3).

10.1 The rank-one and characteristic-p cases

For a p-quasisyntomic ring R let

$$G_i(R) = \operatorname{gr}^i_{\mathrm{N}} \mathbb{\Delta}_R/p, \qquad W_i(R) = \operatorname{fib}(\theta : G_i(R) \longrightarrow G_{i+p}(R)) \quad (i \geq 0).$$

As in Section 4.2, put

$$H_k(R) = \overline{\mathbb{\Delta}}_R\{k\}/p, \qquad \widehat{H}_k(R) = \mathrm{R}\varprojlim_q B_q^k(R), \qquad E_k(R) = \mathrm{cofib}\big(H_k(R) \longrightarrow \widehat{H}_k(R)\big).$$

Here $B_q^k(R)$ denotes the finite quotient for the induced twisted Nygaard filtration [4, Construction 3.3], as in Section 4.2.

We use the following form of the Bhatt–Mathew criterion [4, Proposition 4.3 and Proposition 2.6]: if R is weakly F-smooth and $E_k(R) = 0$ for all $k \in \mathbf{Z}$, then R is F-smooth. Conversely, F-smoothness forces the vanishing of all these completion defects.

Proposition 10.1. *Every p-complete mixed-characteristic rank-one valuation ring is F-smooth. Every characteristic-p valuation ring is F-smooth. In either case $E_k = 0$ for all $k \in \mathbf{Z}$.*

Proof. For a nondiscrete rank-one ring, weak F-smoothness is Theorem 3.6 and $E_k = 0$ is Theorem 9.1, so the preceding criterion applies. A discrete p-complete rank-one ring is a complete DVR and hence F-smooth by [4, Theorem 4.15].

A characteristic-p valuation ring is Cartier smooth by results of Gabber–Ramero and Gabber; see [14, Section 2]. It is therefore F-smooth by [4, Proposition 4.14]. In both cases the completion defects vanish by [4, Proposition 4.3]. □

10.2 Excision for the θ-fibers and the completion defects

Let V be a p-complete valuation ring of mixed characteristic $(0, p)$, and set

$$\mathfrak{q} = \sqrt{pV}, \qquad B = V_{\mathfrak{q}}, \qquad C = V/\mathfrak{q}, \qquad D = \kappa(\mathfrak{q}). \tag{73}$$

We consider the valuation-theoretic Milnor square

$$\begin{array}{ccc} V & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D. \end{array} \tag{74}$$

By [1, Remark 7.24] these are p-complete valuation rings, and [1, Proposition 7.25] proves excision for the Nygaard-complete prismatic pieces and steps on this square.

Lemma 10.2 (Excision)**.** *For every $i \geq 0$ and $k \in \mathbf{Z}$, each of the functors*

$$W_i(-), \quad H_k(-), \quad B_q^k(-), \quad \widehat{H}_k(-), \quad E_k(-)$$

carries the square (74) to a cartesian square (for all $q \geq p$).

Proof. The fibers W_i. The excision of the Nygaard graded pieces follows from [1, Proposition 7.25]: Nygaard completion does not alter a fixed finite associated graded piece, and reduction modulo p is exact. Thus G_i is excisive. Multiplication by θ is natural, so its fiber W_i also satisfies excision.

The Hodge–Tate twists before Nygaard completion. By [1, Lemma 7.22], all relevant derived exterior powers of the cotangent complex satisfy excision on (74). They are the associated graded terms of the conjugate filtration on the p-complete diffracted Hodge complex $\widehat{\Omega}_R^{\mathrm{D}}$ of Bhatt–Lurie [2, Construction 4.7.1]. Finite conjugate stages therefore satisfy excision. After reduction modulo p, the conjugate filtration is exhaustive; filtered colimits commute with finite limits in the stable

category. Consequently $\widehat{\Omega}^{\mathrm{D}}_R/p$ satisfies excision. The Sen eigenspace formula [2, Remark 4.7.5] gives, with the standard sign convention,

$$\overline{\mathbb{\Delta}}_R\{k\} \simeq \mathrm{fib}\Big(\Theta + k : \widehat{\Omega}^{\mathrm{D}}_R \longrightarrow \widehat{\Omega}^{\mathrm{D}}_R\Big).$$

Taking fibers and derived reduction modulo p preserves cartesian squares, proving excision for H_k.

Finite quotients and defects. The graded pieces of the induced twisted Nygaard filtration have the functorial description [4, Construction 3.3]

$$\mathrm{gr}^r_{\mathrm{N}} H_k(R) \simeq \mathrm{cofib}\big(\theta : G_{r-p}(R) \longrightarrow G_r(R)\big), \qquad G_j = 0 \ (j < 0). \tag{75}$$

By the first paragraph, these pieces satisfy excision. Each B^k_q is assembled from finitely many of them by extensions, so finite quotients satisfy excision. Limits of cartesian squares are cartesian, hence $\widehat{H}_k = \mathrm{R}\varprojlim_q B^k_q$ is excisive. Finally E_k is the cofiber of the natural transformation $H_k \to \widehat{H}_k$, and finite limits and colimits agree in the stable category. This proves the lemma. □

10.3 Detecting Tor-amplitude over a valuation ring

We use the following localization criterion for the Tor bounds on W_i.

Lemma 10.3 (Localization and Tor-amplitude over a valuation domain)**.** *Let V be a valuation domain with p a nonzero nonunit, let $S \subset V \setminus pV$ be a multiplicative subset, and let $M \in D(V/p)$. Fix an integer a and assume:*

1. *$S^{-1}M$ has p-complete Tor-amplitude in degrees $\geq a$ over $S^{-1}V$;*
2. *for every $f \in S$, the complex $M \otimes^{\mathbf{L}}_V V/f$ belongs to $D^{\geq a}$.*

Then M has p-complete V-Tor-amplitude in degrees $\geq a$.

Proof. Write $V_S = S^{-1}V$. Since M_S comes from $D(V_S/p)$ and p is a nonzerodivisor in V_S, multiplication by p on M_S is nullhomotopic. Consequently

$$M_S \otimes^{\mathbf{L}}_{V_S} V_S/p \simeq M_S \oplus M_S[1].$$

The first hypothesis, tested on the discrete module V_S/p, therefore forces $M_S \in D^{\geq a+1}$.

For each $f \in S$, the element f is a nonzerodivisor in V. Hence $M \otimes^{\mathbf{L}}_V V/f$ is the cofiber of multiplication by f on M. The second hypothesis and the associated long exact sequence imply that multiplication by f is an isomorphism on $H^t(M)$ for every $t < a$ and an injection on $H^a(M)$. Meanwhile $S^{-1}H^t(M) = H^t(M_S) = 0$ for every $t \leq a$. Thus every class in $H^t(M)$, $t \leq a$, is killed by some $f \in S$; since multiplication by every such f is injective, the class is zero. We conclude that $M \in D^{\geq a+1}$.

Finally, every module over a valuation domain has flat dimension at most one. If N is any discrete V/p-module, resolve it as a V-module by two flat terms. Tensoring that resolution with $M \in D^{\geq a+1}$ gives $M \otimes^{\mathbf{L}}_V N \in D^{\geq a}$. This is the required p-complete Tor-amplitude bound. □

10.4 Valuation rings of arbitrary rank

Proposition 10.4. *Every p-complete valuation ring V of mixed characteristic is F-smooth.*

Proof. By the cotangent complex estimate [11, Theorem 6.5.8] (as used in [1, Proposition 7.25]), V is p-quasisyntomic. Set $\mathfrak{q}$, B, C, D as in (73). Because V is p-adically separated, $\mathfrak{q}$ is its smallest nonzero prime. Indeed, if $\mathfrak{r} \neq (0)$ is prime and $0 \neq x \in \mathfrak{r}$, choose n with $x \notin p^n V$. Comparability of principal ideals gives $p^n V \subset xV$, hence $p \in \mathfrak{r}$ and $\mathfrak{q} \subset \mathfrak{r}$. Thus $\mathfrak{q}$ has height one. Consequently $B = V_{\mathfrak{q}}$ is a p-complete rank-one valuation ring. The rings C and D are valuation rings of characteristic p. All three are F-smooth by Proposition 10.1.

Weak F-smoothness. Fix $i \geq 0$ and put

$$M = W_i(V), \qquad a = i + 1, \qquad S = V \setminus \mathfrak{q}.$$

The localization is relatively perfect modulo p: localization at S and at S^p gives the same characteristic-p algebra. Hence [4, Proposition 3.7] gives

$$S^{-1}M \simeq W_i(B). \tag{76}$$

Since B is F-smooth, the right side has p-complete B-Tor amplitude in degrees $\geq i + 1$.

Let $f \in S$. By Lemma 10.2, the valuation square gives a fiber sequence

$$M \longrightarrow W_i(B) \oplus W_i(C) \longrightarrow W_i(D).$$

The element f is invertible in B and in D, so tensoring with V/f yields

$$M \otimes^{\mathbf{L}}_V V/f \simeq W_i(C) \otimes^{\mathbf{L}}_V V/f.$$

For any $x \in \mathfrak{q}$, comparability of xV with fV and $f \notin \mathfrak{q}$ gives $xV \subset fV$; therefore $\mathfrak{q} \subset fV$ and $V/f \simeq C/\overline{f}$. Moreover, $\overline{f} \neq 0$ is a nonzerodivisor in the valuation domain C, so the two-term resolution of V/f identifies $C \otimes^{\mathbf{L}}_V V/f \simeq C/\overline{f}$. We obtain

$$M \otimes^{\mathbf{L}}_V V/f \simeq W_i(C) \otimes^{\mathbf{L}}_C C/\overline{f} \in D^{\geq i+1}, \tag{77}$$

where the last bound follows from the weak F-smoothness of C. Apply Lemma 10.3 to (76) and (77). It follows that $W_i(V)$ has p-complete V-Tor-amplitude in degrees $\geq i + 1$. As i was arbitrary, V is weakly F-smooth.

Nygaard completeness. For any $k \in \mathbf{Z}$, Lemma 10.2 gives a fiber sequence

$$E_k(V) \longrightarrow E_k(B) \oplus E_k(C) \longrightarrow E_k(D).$$

The three terms on the right vanish by Proposition 10.1 and the Bhatt–Mathew criterion. Thus $E_k(V) = 0$ for every k. Weak F-smoothness together with vanishing of the completion defects proves the claim by [4, Proposition 4.3]. □

10.5 Removing the p-completeness hypothesis

Lemma 10.5. *Let V be a valuation ring in which p is a nonzero nonunit. Its ordinary p-adic completion $\widehat{V} = \varprojlim_n V/p^n$ is a p-torsion-free, p-complete valuation ring.*

Proof. The inverse limit of the local rings V/p^n is local. We first verify

$$\ker(\widehat{V} \longrightarrow V/p^n) = p^n \widehat{V}. \tag{78}$$

Indeed, a compatible system in the kernel can be divided modulo p^m $(m \geq n)$ by p^n to obtain a unique element modulo p^{m-n}, because p is a nonzerodivisor in V. The divisions are compatible and define an element of $\widehat{V}$. The opposite inclusion is immediate.

Take $0 \neq x \in \widehat{V}$. Choose n and $a \in V \setminus p^n V$ representing the nonzero image of x in V/p^n. Since principal ideals are comparable, $p^n V \subset aV$, so $p^n = ac$ for some nonunit $c \in V$. By (78),

$$x = a + p^n z = a(1 + cz), \qquad z \in \widehat{V}.$$

The element c belongs to the maximal ideal of $\widehat{V}$, hence $1 + cz$ is a unit. Every nonzero element of $\widehat{V}$ is therefore associated to the image of an element of V, which shows that its nonzero principal ideals are linearly ordered.

These elements are not zero divisors. Indeed, any nonzero image $a \in V$ divides some p^N, and similarly $b \mid p^M$. If ab had zero image in $\widehat{V}$, it would belong to every $p^r V$, contradicting $ab \mid p^{N+M}$ (take $r = N + M + 1$). Thus $\widehat{V}$ is a valuation domain. Multiplication by p is injective, and p-completeness is built into its construction. □

Proof of Theorem 1.2. If p is invertible in V, its derived p-completion is zero and the F-smoothness condition is vacuous. If V has characteristic p, use Proposition 10.1. Otherwise V is a valuation domain of mixed characteristic $(0, p)$ with p a nonunit. Since V is p-torsion-free, its ordinary p-adic completion agrees with its derived p-completion. By Lemma 10.5 this completion is again a valuation ring, and by Proposition 10.4 it is F-smooth. By [11, Theorem 6.5.8], V is p-quasisyntomic. Since F-smoothness depends only on the p-completion [4, Definition 4.1], V is F-smooth. □

Remark 10.6. Theorem 1.2 does not follow formally from Theorem 1.1. Its proof uses the rank-one results directly, and needs excision for E_k and the Tor bound for W_i separately.

11 Proof of Theorem 1.1

Proposition 11.1 (Complete rank-one case)**.** *For every complete rank-one valuation ring V of mixed characteristic $(0, p)$ and every $n \geq 0$,*

$$\operatorname{cofib}(S_n(V) \longrightarrow E_n(V)) \in D^{\geq n}(\mathbf{F}_p).$$

Proof. For $n = 0$, both complexes are connective and the comparison induces the identity on $H^0 = \mathbf{F}_p$; see [2, Section 8.4]. Let $n \geq 1$, and suppose first that V is nondiscrete. By Theorem 9.1, all completion defects vanish, so Proposition 4.4 gives

$$\tau_{\leq j-1} C_j(V) = 0 \qquad (j \geq 1).$$

Since C_j is the cofiber of $v_1 : S_j \to S_{j+p-1}$, multiplication by v_1 is injective on $H^q S_j(V)$ for $q \leq j$. Applying this along the Bott tower starting in weight n, and using the localization equivalence (71), we see that

$$H^q S_n(V) \longrightarrow H^q E_n(V)$$

is injective for $q \leq n$. Proposition 2.6 gives surjectivity for $0 \leq q < n$. For $q < 0$ the target vanishes, so injectivity shows that the source vanishes too.

If V is discrete, it is a complete discrete valuation ring, and the assertion follows from the comparison theorem for regular rings [4, Theorem 4.15 and Proposition 5.2]. □

Proof of Theorem 1.1. Combine Proposition 11.1 with Proposition 2.1. □

12 Applications

We record some consequences of Theorems 1.1 and 1.2, obtained by combining them with known results. Throughout this section, $\mathbf{Z}(i)_{\mathrm{mot}}$ denotes the (not generally $\mathbb{A}^1$-invariant) motivic complex of Elmanto–Morrow and Bouis; $\mathbf{Z}(i)_{\mathrm{cdh}}$ denotes cdh-motivic cohomology; and $\mathbf{Z}(i)_{\mathbb{A}^1}$ denotes the $\mathbb{A}^1$-invariant motivic complex of Bachmann–Elmanto–Morrow.

12.1 Consequences of F-smoothness

Corollary 12.1 (Stability under smooth extensions)**.** *Let V be a valuation ring and fix a prime p. Polynomial and étale V-algebras are F-smooth at p, as are smooth V-algebras satisfying the standard p-quasisyntomic hypotheses. The same is true for filtered colimits of such F-smooth rings.*

Proof. By Theorem 1.2, V is F-smooth. Now apply [4, Propositions 4.6–4.9], which give stability under filtered colimits, étale maps, polynomial extensions, and p-completely flat extensions that are smooth modulo p. □

By Theorem 1.2, the description of prismatic Frobenius in terms of the Beilinson t-structure [4, Remark 4.11] applies to every valuation ring.

12.2 cdh-motivic cohomology on arbitrary schemes

Theorem 12.2 (cdh-local F-smoothness)**.** *Every commutative ring is cdh-locally F-smooth in the sense of [10, Definition 6.4].*

Proof. The defining condition is that every valuation ring over the given ring be F-smooth at all primes, which is Theorem 1.2. □

Corollary 12.3 (cdh comparison and projective bundles)**.** *For any quasi-compact quasi-separated scheme X and $i \geq 0$, the canonical comparison is an equivalence*

$$\mathbf{Z}(i)_{\mathrm{cdh}}(X) \xrightarrow{\sim} \mathbf{Z}(i)_{\mathbb{A}^1}(X).$$

The cdh-motivic complexes are therefore $\mathbb{A}^1$-invariant on all such schemes and satisfy the projective bundle formula. More precisely, for a rank-d vector bundle $\mathcal{E}$ on X, the usual first-Chern-class map gives

$$\bigoplus_{r=0}^{d-1} \mathbf{Z}(i-r)_{\mathrm{cdh}}(X)[-2r] \xrightarrow{\sim} \mathbf{Z}(i)_{\mathrm{cdh}}(\mathbf{P}(\mathcal{E})),$$

with the customary convention for negative weights.

Proof. Apply [10, Theorem 6.7] to the base ring $\mathbf{Z}$, which is cdh-locally F-smooth by Theorem 12.2. The identification with the $\mathbb{A}^1$-invariant theory and its projective bundle formula are established there using [1, Theorems 1.10 and 8.20]. □

12.3 A global comparison bound

Theorem 12.4 (Global mixed-characteristic comparison)**.** *For every quasi-compact quasi-separated scheme X, every prime p, and every $i \geq 0$, the cofiber of the canonical comparison*

$$\mathbf{Z}(i)_{\mathbb{A}^1}(X)/p \longrightarrow R\Gamma_{\acute{e}t}(X[1/p], \mu_p^{\otimes i})$$

belongs to $D^{\geq i-1}(\mathbf{F}_p)$. By Corollary 12.3, the same holds with $\mathbf{Z}(i)_{\mathrm{cdh}}$ in place of $\mathbf{Z}(i)_{\mathbb{A}^1}$.

Proof. Theorem 1.1 verifies the hypothesis $\mathrm{Val}(\operatorname{Spec}\mathbf{Z}, p, j)$ for every j. Apply [1, Theorem 1.13] with base $S = \operatorname{Spec}\mathbf{Z}$, which has finite valuative dimension. The cdh replacement follows from Corollary 12.3. □

Remark 12.5. Theorem 12.4 concerns $\mathbb{A}^1$-invariant motivic cohomology; it says nothing about $\mathbf{Z}(i)_{\mathrm{mot}}$ over general regular rings.

12.4 Motivic homotopy invariance over Prüfer domains

Theorem 12.6 (Integral motivic homotopy invariance)**.** *Let B be a Prüfer domain, let X be an ind-smooth B-scheme, and let $m \geq 1$. For every $i \geq 0$ the natural map*

$$\mathbf{Z}(i)_{\mathrm{mot}}(X) \xrightarrow{\sim} \mathbf{Z}(i)_{\mathrm{mot}}(\mathbb{A}^m_X)$$

is an equivalence in $D(\mathbf{Z})$. On this class of schemes the canonical maps also induce equivalences

$$\mathbf{Z}(i)_{\mathrm{mot}}(X) \simeq \mathbf{Z}(i)_{\mathrm{cdh}}(X) \simeq \mathbf{Z}(i)_{\mathbb{A}^1}(X).$$

Proof. Theorem 12.2 makes B cdh-locally F-smooth. Apply [10, Theorem 6.9 and its proof]; the last equivalence is Corollary 12.3. □

12.5 Finite-coefficient motivic cohomology over valuation rings

Theorem 12.7 (Generic-fiber calculation)**.** *Let V be a p-torsion-free valuation ring and R a henselian local V-algebra that is ind-smooth over V. For integers $i, n \geq 0$ and $r \geq 1$ there are natural identifications*

$$H^n\big(R, \mathbf{Z}/p^r(i)_{\mathrm{mot}}\big) \simeq \begin{cases} H^n_{\acute{e}t}(R[1/p], \mu_{p^r}^{\otimes i}), & n < i, \\ \operatorname{im}\Big((R^\times)^{\otimes i} \longrightarrow H^i_{\acute{e}t}(R[1/p], \mu_{p^r}^{\otimes i})\Big), & n = i, \\ 0, & n > i. \end{cases}$$

Here the arrow in the middle line is the étale symbol map.

Proof. Since V is F-smooth by Theorem 1.2, this is [10, Theorem C], which uses [4, Theorem 1.8] and the Beilinson–Lichtenbaum comparison [10, Theorem 5.1] for ind-smooth algebras over valuation rings. □

Corollary 12.8 (Improved Milnor K-theory)**.** *Let V be a henselian p-torsion-free valuation ring, and let $i \geq 0$ and $r \geq 1$. The Galois symbol gives an injection*

$$\widehat{K}^M_i(V)/p^r \hookrightarrow H^i_{\acute{e}t}(V[1/p], \mu_{p^r}^{\otimes i}),$$

whose image is the subgroup generated by symbols of units of V. If p is a nonzero nonunit and V has rank one, then $V[1/p] = \mathrm{Frac}(V)$.

Proof. Apply Theorem 12.7 with $R = V$ and $n = i$. By [9, Theorem 2.21], the improved Milnor group modulo p^r identifies with $H^i(V, \mathbf{Z}(i)_{\mathrm{mot}})/p^r$. Furthermore, [8, Theorem 5.1] proves that the integral motivic complex of a henselian valuation ring is concentrated in degrees at most i. The coefficient sequence therefore identifies this quotient with $H^i(V, \mathbf{Z}/p^r(i)_{\mathrm{mot}})$, so the corollary follows. □

Remark 12.9 (Syntomic nearby cycles). For a p-torsion-free F-smooth scheme X and $j : X[1/p] \hookrightarrow X$, [4, Theorem 1.8] identifies $\mathbf{Z}/p^r(i)_{\mathrm{syn},X} \to \tau_{\leq i} Rj_*\mu_{p^r}^{\otimes i}$ in cohomological degrees below i, and makes the map on H^i injective with image generated by unit symbols. Corollary 12.1 supplies this for smooth schemes over a p-torsion-free valuation base when the hypotheses of that theorem hold.

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